

Neural-Adaptive Finite-Time Consensus Tracking of Singular Multiagent Systems With Actuator Attack

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Abstract—Since the actuator attack is an important threat to the complicated network, the problem of finite-time consensus tracking (FTCT) control for time-varying delay singular multiagent systems with actuator attack is investigated in this article. First, to approximate the states of the follower agents and design the sliding mode surfaces under actuator attack, a state observer is proposed. Then, the novel neural-adaptive distributed FTCT protocol and actuator attack estimation protocol are designed based on the radial basis function neural network estimation method. The sliding mode control with a partitioning strategy is adopted for the FTCT problem analysis of singular multiagent systems. The sufficient conditions and solving strategies of FTCT for singular multiagent systems over the whole finite-time interval are provided. Finally, the validity of FTCT for singular multiagent systems is verified by simulation examples.

Index Terms—Finite-time consensus tracking (FTCT), neural-adaptive distributed protocol, singular multiagent systems, sliding mode control.

NOMENCLATURE

$\mathcal{G}$	Undirected topology graph.
$\mathcal{V}$	Node set in $\mathcal{G}$.
$\mathcal{E}$	Edge set in $\mathcal{G}$.
$\mathbb{F}$	Set of follower agents.
N	Number of follower agents.
$\mathcal{A} \in \mathbb{R}^{N \times N}$	Adjacent matrix.
$\mathcal{L}$	Laplacian matrix.
$\lambda_{\max}(A)$	Maximum eigenvalue of A .
$\lambda_{\min}(A)$	Minimum eigenvalue of A .
$\text{tr}(A)$	Trace of the matrix A .
$\ A\ $	Euclidean norm of A .
$\ A\ _F$	Frobenius norm of A .
$\mathbb{I}_{[0,T],\varpi}$	Set of integrals bounded disturbances within limited time T .
$\text{sym}(A)$	Sum of A and its transpose A^T .
$\text{sgn}\{x\}$	Sign function of x .
$\Delta_i \in \mathbb{R}^{l_i \times p}$	Connection weights from hidden layer to output layer.

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$\Pi_i \in \mathbb{R}^{(2n+1) \times l_i}$ Connection weights from input layer to hidden layer.

I. INTRODUCTION

THE cooperative control problem for multiagent systems is the hot spot in the artificial intelligence research and has received great attention from scholars [1], [2], [3]. And it has been widely applied in many fields, such as autonomous aerial vehicles, disaster relief, and multirobot collaboration, etc. Consensus control is the foundation of cooperative control research [4], [5], which can be subdivided into consensus tracking, formation, flocking, etc. Up until now, the consensus control based on the Lyapunov asymptotic stability is still the main research direction [6], [7], [9]. However, the multiagent systems are required to achieve the control target in a finite time for some practical applications.

Compared with asymptotic stability consensus control, finite-time consensus control has high precision, high reliability and strong robustness, and has become a hot topic in consensus control research recently [8], [10], [11], [12], [13], [14], [15], [16], [17]. In [10], [11], and [12], the distributed protocols based on the sign function are built to drive the multiagent to achieve consensus within a finite time. The nonfragile consensus protocol is proposed based on the sliding mode control that guarantees the finite-time stability (FTS) of the multiagent systems with time-varying delay in [13]. The finite-time consensus tracking (FTCT) problem of heterogenous multiagent systems with uncertainties and external disturbances is solved by using the sliding mode control method [14]. To further accurately analyze the state trajectories of the reaching phase and sliding motion phase of the FTS based on sliding mode control, the partitioning strategy is proposed in [15]. The FTS sufficient conditions of the UMV system with DoS attacks are proposed based on the sliding mode control with a partitioning strategy [16]. For the linear multiagent systems, the FTCT problem is solved by the partitioning strategy and genetic algorithm [17]. Different from the finite-time cooperative control based on the sign function, the sliding mode control has stronger robustness to external attacks and disturbances. The sliding mode control with partitioning strategy is an effective method to solve finite-time control problems, which has inspired the research of this article.

It is worth noting that the above research results assume that the state dimension is equal to the dynamic order, that is, there is no algebraic constraints on the states of multiagent

systems. In order to more accurately describe some complex dynamics models (circuit systems, constrained robotic systems, aircraft systems, etc.), the singular systems which contain algebraic constraints are adopted [18], [19], [20]. And the cooperative control for singular multiagent systems has been widely studied in recent years. For the linear singular multiagent systems, the outputs-based distributed consensus protocol is designed to achieve admissible consensus in [21]. In [22], the sufficient consensus results related to eigenvalues of switching topologies are constructed for singular multiagent systems. The communication network between singular multiagent systems is a signed digraph, and the fully distributed formation protocol is constructed for the formation tracking analysis [23]. It should be pointed out that the actuator attack and external disturbances that will affect and even threaten the cooperative control of the system have not been investigated in the above research results. Therefore, it is necessary and challenging to design a protocol that can suppress the effects caused by actuator attack and external disturbances in the practical application.

The actuator attack of system is one of the major threats to the security performance of multiagent systems. The radial basis function neural network can estimate a nonlinear function accurately, which has been widely adopted in the cooperative control analysis of multiagent systems [24], [25], [26], [27]. In [24], the adaptive fault-tolerant protocol is designed based on the radial basis function neural network, which can guarantee the FTS of the nonlinear multiagent systems. The neural network-based consensus tracking protocol is designed which can effectively suppress actuator failures and drive the nonlinear multiagent systems to achieve FTCT in [25]. Furthermore, the sufficient conditions for the scaled consensus tracking problem are presented, where external disturbance and actuator attack of singular multiagent systems are well suppressed by the designed neural-adaptive protocol [26]. The FTCT of singular multiagent systems can describe the transient performance of the system more accurately, which has not been fully considered in the previous research results. The algebraic constraints of states in singular dynamics make the signed function-based FTS protocol design method of normal multiagent systems invalid [10], [11], [12]. And the FTS control results of normal multiagent systems based on the sliding mode control in [13], [14], [15], [16], and [17] cannot directly deal with the FTCT problem of singular multiagent systems. Therefore, the FTCT control of time-varying delay singular multiagent systems with unknown actuator attack and external disturbances is an interesting direction that deserves to be solved.

Based on the above analysis, the prime objective of this article is to find the design method of the FTCT protocol such that the time-varying delay singular follower agents can track the leader agent in a finite time under actuator attack and external disturbances. There are three challenges when using sliding mode control with a partitioning strategy to solve the FTCT control problem in this article. First, accurately estimating and attenuating actuator attacks related to the current state and delay state is a challenge. Second, the sliding mode surfaces based on system state are susceptible

to actuator attacks, so designing a suitable reachability sliding mode surface is a challenge. Third, how to design a distributed protocol to drive the state of agents to reach consensus tracking within a finite time is a challenge. The main innovation points of this work are condensed as follows.

- 1) The FTCT control of time-varying delay singular multiagent systems is investigated for the first time. Compared with the cooperative results based on asymptotic stability in [21], [22], [23], and [26], FTCT has high precision, high dependability, and strong countermeasure ability.
- 2) The actuator attacks are accurately estimated and attenuated by the designed state observer and time-varying delay neural network estimation function.
- 3) The sliding mode function based on state error between observer state and leader agent state is designed, which is insensitive to actuator attacks.
- 4) The novel neural-adaptive distributed protocol is designed, which can estimate and suppress actuator attacks through updating radial basis function neural network parameters, and drive agents to achieve FTCT.

Notation: The adjacency matrix $\mathcal{A} = [a_{ij}]$ satisfies $a_{ij} > 0$ (i.e., $(i, j) \in \mathcal{E}$) and $a_{ij} = 0$ (i.e., $(i, j) \notin \mathcal{E}$). The Laplacian matrix $\mathcal{L}$ is defined as $l_{ii} = \sum_{j \neq i} |a_{ij}|$ and $l_{ij} = -a_{ij}$ for all $i \neq j$. The $\mathcal{J} = \text{diag}\{J_1, J_2, \dots, J_N\}$, $J_i > 0$ when leader agent has information exchange with the follower agent i , else $J_i = 0$. The vector $\text{col}\{x_1(t), \dots, x_N(t)\} = [x_1^T(t), \dots, x_N^T(t)]^T$. The main notations used in this article are listed in Nomenclature.

II. PROBLEM FORMULATION

The dynamic of singular multiagent systems with nonlinear actuator attack and time-varying delay is given as follows:

$$\begin{aligned} E\dot{x}_i(t) &= \mathcal{A}_1 x_i(t) + \mathcal{A}_2 x_i(t - d(t)) + \mathcal{B}(u_i(t) \\ &\quad + f_i(x_i(t), x_i(t - d(t)), t)) + \mathcal{D}w_i(t) \quad i \in \mathbb{F} \\ E\dot{x}_0(t) &= \mathcal{A}_1 x_0(t) + \mathcal{A}_2 x_0(t - d(t)) \\ x_i(t) &= \phi_i(t) \quad i \in \mathbb{F} \cup \{0\} \quad t \in [-d, 0] \end{aligned} \quad (1)$$

where $x_i \in \mathbb{R}^n$, $i \in \mathbb{F}$ is the i th follower agent state, $x_0 \in \mathbb{R}^n$ is the leader agent state, $u_i \in \mathbb{R}^p$ is control input, and $w_i \in \mathbb{R}^q$ is external disturbance for $i \in \mathbb{F}$. The time-varying delay of multiagent system is $d(t)$, which satisfies

$$0 \leq d(t) \leq d < \infty, \quad \dot{d}(t) \leq \varepsilon < 1. \quad (2)$$

The nonlinear function $f_i(x_i(t), x_i(t - d(t)), t)$ indicates an attack on the actuator, which will threaten the system's security. For convenience, $f_i(x_i(t), x_i(t - d(t)), t)$ is denoted by f_i in this article. For all agents $i \in \mathbb{F} \cup \{0\}$, the initial condition is $\phi_i(t)$. The $E, \mathcal{A}_1 \in \mathbb{R}^{n \times n}$, $\mathcal{A}_2 \in \mathbb{R}^{n \times n}$, $\mathcal{B} \in \mathbb{R}^{n \times p}$ and $\mathcal{D} \in \mathbb{R}^{n \times q}$ are known system matrices. The matrix E may be singular (i.e., $\text{rank}(E) = r < n$) and $(E, \mathcal{A}_1)$ is regular. Matrix $\mathcal{B}$ is assumed to be of full rank.

For a given time point T , the external disturbance $w_i(t)$ satisfies

$$\mathbb{I}_{[0, T], \varpi} \triangleq \left\{ w_i(t) \in \mathcal{L}_2[0, T] : \int_0^T w_i^T(s) w_i(s) ds \leq \varpi \right\} \quad (3)$$

where $\varpi > 0$ is a known constant.

Based on the neural network estimation method, the nonlinear attack function f_i can be modeled as

$$f_i = \Delta_i^T \sigma_i(\Pi_i^T x_{di}) + \gamma_i(\bar{x}_{di}). \quad (4)$$

The input vector of neural network is $x_{di} = \text{col}\{x_i(t), x_i(t-d(t)), -1\}$, with a bias of -1 . For vector $\bar{x}_{di} = \text{col}\{x_i(t), x_i(t-d(t))\}$ and for any arbitrary constant $\gamma_i > 0$, the $\|\gamma_i(\bar{x}_{di})\| \leq \gamma_i$ is true.

The function $\sigma_i(\cdot)$ can be written as $\sigma_i(\cdot) = \text{col}\{\sigma_{i1}(\cdot), \sigma_{i2}(\cdot), \dots, \sigma_{i l_i}(\cdot)\}$ and

$$\sigma_{ij}(s) = \frac{1}{1 + \exp(-\alpha_{ij}s)}$$

where $\alpha_{ij} > 0$ is the constant.

Remark 1: The actuator of agents may be attacked in some complex environments, and these actuator attacks will threaten system security [28], [29]. For the multiagent systems in [28] and [29], the nonlinear actuator attack is $\delta_a(t, x(t))$ only related to the current state $x_i(t)$. The singular dynamics with time-varying delays are adopted in this article, so the actuator attacks may be related to delay states. Therefore, the delay-independent actuator attacks mentioned above are invalid, and the actuator attack f_i based on the current state $x_i(t)$ and the delay state $x_i(t-d(t))$ is constructed. From (4), the actuator attack can be accurately estimated based on the neural network method.

For $i \in \mathbb{F}$, the estimation matrices Δ_i and Π_i are defined as $\hat{\Delta}_i$ and $\hat{\Pi}_i$, respectively. Then, the estimation error matrices are $\tilde{\Delta}_i = \hat{\Delta}_i - \Delta_i$ and $\tilde{\Pi}_i = \hat{\Pi}_i - \Pi_i$. In addition, $\tilde{f}_i = \hat{\Delta}_i^T \sigma_i(\hat{\Pi}_i^T x_{di})$ is the estimation of nonlinear function f_i .

Lemma 1: [31]: The error function $\tilde{f}_i = \hat{f}_i - f_i$ has the following form:

$$\tilde{f}_i = \tilde{\Delta}_i^T (\hat{\sigma}_i - \hat{\sigma}_i' \hat{\Pi}_i^T x_{di}) + \tilde{\Delta}_i^T \hat{\sigma}_i' \tilde{\Pi}_i^T x_{di} + \delta_i(\bar{x}_{di}). \quad (5)$$

For the residual error $\delta_i(\bar{x}_{di})$, one has

$$\delta_i(\bar{x}_{di}) = \tilde{\Delta}_i^T \hat{\sigma}_i' \Pi_i^T x_{di} + \tilde{\Delta}_i^T o(\tilde{\Pi}_i^T x_{di}) - \varepsilon_i(\bar{x}_{di})$$

where $\hat{\sigma}_i = \sigma_i(\hat{\Pi}_i^T x_{di})$, $\hat{\sigma}_i' = \sigma_i'(\hat{\Pi}_i^T x_{di})$ is Jacobian matrix, and $o(\tilde{\Pi}_i^T x_{di})$ is an infinitesimal term.

Lemma 2: [31]: For the residual error function $\delta_i(\bar{x}_{di})$, the following inequality is true:

$$\|\delta_i(\bar{x}_{di})\| \leq \theta_i^T x_{di}^F \quad (6)$$

where $\theta_i \in \mathbb{R}^4$ is an unknown vector, x_{di}^F is defined as $x_{di}^F = \text{col}\{1, \|x_{di}\|, \|\hat{\Delta}_i^T\|_F, \|\hat{\Pi}_i^T\|_F\}$ and $\tilde{\theta}_i = \hat{\theta}_i - \theta_i$.

Assumption 1: The connection topology of multiagent systems (1) contains a spanning tree. And the root node of the spanning tree is the leader agent.

Lemma 3: [30]: If Assumption 1 is hold, the real parts of the eigenvalues of $\mathcal{L} + \mathcal{J}$ are all positive.

Lemma 4: Given the symmetric positive matrix $Y \in \mathbb{R}^{n \times n}$ and any matrix $X \in \mathbb{R}^{n \times s}$, the following inequality holds:

$$\lambda_{\min}(Y) \text{tr}(X^T X) \leq \text{tr}(X^T Y X) \leq \lambda_{\max}(Y) \text{tr}(X^T X).$$

Proof: For the matrix Y , an orthogonal matrix G satisfying

$$G^T Y G = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_N\}$$

can be found. Defining $G^{-1}X = [g_{ij}]_{n \times s}$, the following equation can be obtained:

$$\begin{aligned} \text{tr}(X^T Y X) &= \text{tr}\left((G^{-1}X)^T \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_N\} G^{-1}X\right) \\ &= \text{tr} \begin{bmatrix} \sum_{i=1}^n \lambda_i g_{i1}^2 & * & \dots & * \\ * & \sum_{i=1}^n \lambda_i g_{i2}^2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & \sum_{i=1}^n \lambda_i g_{is}^2 \end{bmatrix} \\ &= \sum_{j=1}^s \sum_{i=1}^n \lambda_i g_{ij}^2 \end{aligned}$$

where $*$ are not involved in the proof and are omitted.

For the orthogonal matrix G , it can be obtained that

$$\begin{aligned} \lambda_{\min}(Y) \text{tr}(X^T X) &= \lambda_{\min}(Y) \text{tr}\left((G^{-1}X)^T G^{-1}X\right) \\ &= \lambda_{\min}(Y) \text{tr} \begin{bmatrix} \sum_{i=1}^n g_{i1}^2 & * & \dots & * \\ * & \sum_{i=1}^n g_{i2}^2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & \sum_{i=1}^n g_{is}^2 \end{bmatrix} \\ &\leq \sum_{j=1}^s \sum_{i=1}^n \lambda_i g_{ij}^2 \\ &= \lambda_{\max}(Y) \text{tr} \begin{bmatrix} \sum_{i=1}^n g_{i1}^2 & * & \dots & * \\ * & \sum_{i=1}^n g_{i2}^2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & \sum_{i=1}^n g_{is}^2 \end{bmatrix} \\ &= \lambda_{\max}(Y) \text{tr}\left((G^{-1}X)^T G^{-1}X\right) \\ &= \lambda_{\max}(Y) \text{tr}(X^T X). \end{aligned}$$

The proof is completed. $\blacksquare$

Definition 1: [18]: The pair $(E, \mathcal{A})$ is impulse free when $\text{degdet}(sE - \mathcal{A}) = \text{rank}(E)$. The pair $(E, \mathcal{A})$ is admissible if $(E, \mathcal{A})$ is impulse-free and stable.

For each follower agent i in (1), the state observer is designed as follows:

$$\begin{aligned} E\dot{\eta}_i(t) &= \mathcal{A}_1 \eta_i(t) + \mathcal{A}_2 \eta_i(t-d(t)) + \mathcal{B}(u_i(t) + u_{fi}(t)) \\ \eta_i(t) &= \hat{\phi}_i(t) \quad t \in [-d, 0] \quad i \in \mathbb{F} \end{aligned} \quad (7)$$

where $\eta_i(t), \eta_i(t-d(t)) \in \mathbb{R}^n$ are the state of observer system, $u_{fi}(t)$ can attenuate the actuator attack caused by f_i . The $\hat{\phi}_i(t)$ is the initial condition.

According to the singular multiagent systems (1), it can be calculated that

$$E\dot{r}_i(t) = \mathcal{A}_1 r_i(t) + \mathcal{A}_2 r_i(t-d(t)) + \mathcal{B}(u_i(t) + f_i) + \mathcal{D}w_i(t) \quad (8)$$

where $r_i(\cdot) = x_i(\cdot) - x_0(\cdot)$.

Define $r(t) = \text{col}\{r_1(t), r_2(t), \dots, r_N(t)\}$. The definition of FTCT for singular multiagent systems (1) is proposed in the following.

Definition 2: Given a time interval $[0, T]$, the scalars $l_2 > l_1 > 0$ and $\mathcal{R} > 0$, if the closed-loop system is impulse free and satisfies

$$\begin{aligned} r^T(0) (I_N \otimes E^T \mathcal{R} E) r(0) &\leq l_1 \\ \Rightarrow r^T(t) (I_N \otimes E^T \mathcal{R} E) r(t) &\leq l_2 \quad \forall t \in [0, T] \end{aligned}$$

for all $w_i \in \mathbb{I}_{[0, T], \omega}$. Then, the system state $r(t)$ is FTS with respect to $(l_1, l_2, [0, T], I_N \otimes \mathcal{R}, \mathbb{I}_{[0, T]})$. And the singular multiagent systems (1) is FTCT with respect to $(l_1, l_2, [0, T], I_N \otimes \mathcal{R}, \mathbb{I}_{[0, T]})$.

The items in $(l_1, l_2, [0, T], I_N \otimes \mathcal{R}, \mathbb{I}_{[0, T], \omega})$ correspond to the key metrics for FTCT: l_1 is the initial states of $r^T(t)(I_N \otimes E^T \mathcal{R} E)r(t)$, l_2 is upper bounded of $r^T(t)(I_N \otimes E^T \mathcal{R} E)r(t)$, T is the control time, $I_N \otimes \mathcal{R}$ is parameter matrix and $\mathbb{I}_{[0, T], \omega}$ is the set of bounded external disturbances. Since $I_N \otimes \mathcal{R}$ and $\mathbb{I}_{[0, T]}$ remain unchanged, $(l_1, l_2, [0, T], I_N \otimes \mathcal{R}, \mathbb{I}_{[0, T], \omega})$ is simply denoted as $(l_1, l_2, [0, T])$ in this article.

Remark 2: The definition of FTCT with respect to $(l_1, l_2, [0, T])$ for singular multiagent systems is proposed in this article. Recently, the consensus tracking problem of singular multiagent systems has been investigated [20], [26], [27]. In [20], the issue of consensus tracking for singular multiagent systems fails to address time-varying delays and actuator attacks. Although [26], [27] consider the effects on time-varying delays and actuators, their methods are nevertheless based on asymptotic stability with infinite convergence time and inherently unable to meet the actual requirements for finite-time control accuracy. In [8], [10], and [11], the terminal sliding mode control method and adaptive neural networks method were employed to achieve finite-time consensus control for normal multiagent systems. However, the aforementioned finite-time consensus control results are all based on an autonomous system $\dot{x} = f(x)$ without algebraic constraints. This cannot deal with the FTS analysis of $E\dot{x} = f(x)$ when $\text{rank}(E) = r < n$. This stems from the fact that the solutions of the singular system include both impulse terms and the derivatives of both the input and disturbance terms. Therefore, the sliding mode-based finite-time analysis method is introduced for the FTCT problem analysis in this article.

Remark 3: The sliding mode control can be divided into the reaching phase and the sliding motion phase. For $0 < T_r \leq T$, assuming interval $[0, T_r]$ is the reaching process and interval $[T_r, T]$ is the sliding motion process. By analyzing the state trajectories of two stages, the finite time cooperative control problem of normal multiagent systems is resolved [15], [16], [17]. However, the singular dynamics of the multiagent systems, time-varying delays, and the unknown actuator attacks make the study of this article more challenging and interesting.

Definition 3: For interval $[0, T]$ and $0 < l_1 < l_1^* < l_2$, the singular multiagent system (1) is said to be FTCT with respect to $(l_1, l_2, [0, T])$, if it is FTCT with respect to $(l_1, l_1^*, [0, T_r])$ in reaching phase and FTCT with respect to $(l_1^*, l_2, [T_r, T])$ in sliding motion phase.

III. MAIN RESULTS

In this section, the main purpose is to design a distributed protocol based on sliding mode control method to achieve FTCT of reaching phase $[0, T_r]$ and sliding motion phase $[T_r, T]$.

A. Distributed Adaptive Protocol Design and Reaching Analysis

In this subsection, the main objective is to find a neural-adaptive FTCT protocol design method that enables the error system to be driven into the sliding mode surface in a finite time.

The relative state error between observer state and leader agent state is $\tilde{r}_i(\cdot) = \eta_i(\cdot) - x_0(\cdot)$. From (1) and (7), the following error system can be obtained:

$$E\dot{\tilde{r}}_i(t) = \mathcal{A}_1 \tilde{r}_i(t) + \mathcal{A}_2 \tilde{r}_i(t - d(t)) + \mathcal{B} [u_i(t) + u_{fi}(t)]. \quad (9)$$

The integral sliding mode surfaces are selected as

$$\begin{aligned} s_i(t) &= \mathcal{B}^T \mathcal{X} E \tilde{r}_i(t) - \mathcal{B}^T \mathcal{X} \int_0^t [\mathcal{A}_1 \tilde{r}_i(\tau) + \mathcal{A}_2 \tilde{r}_i(\tau - d(\tau)) \\ &\quad + \mathcal{B} \mathcal{K} \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\tilde{r}_j(\tau) - \tilde{r}_i(\tau)) - J_i \tilde{r}_i(\tau) \right)] d\tau \end{aligned} \quad (10)$$

where $\mathcal{X} \in \mathbb{R}^{n \times n}$ and $\mathcal{K} \in \mathbb{R}^{p \times n}$ are the matrices that need to be designed.

The sliding mode-based neural-adaptive FTCT protocol is constructed as

$$\begin{aligned} u_{fi}(t) &= \hat{\Delta}_i^T \sigma_i (\hat{\Pi}_i^T x_{di}) + \hat{\theta}_i^T x_{di}^F \text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r_i(t)) \\ u_i(t) &= \mathcal{K} \sum_{j \in \mathcal{N}_i} [a_{ij} (\tilde{r}_j(t) - \tilde{r}_i(t)) - J_i \tilde{r}_i(t)] - u_{fi}(t) \\ &\quad - \rho_i \text{sgn}_p(s_i(t)) \end{aligned} \quad (11)$$

and the neural-adaptive rules are as follows:

$$\begin{aligned} \dot{\hat{\Delta}}_i &= \mathcal{P}_{1i} (\hat{\sigma}_i - \hat{\sigma}_i' \hat{\Pi}_i^T x_{di}) \mathcal{B}^T \mathcal{X}_1^T r_i(t) \\ \dot{\hat{\Pi}}_i &= \mathcal{P}_{2i} x_{di} (\mathcal{B}^T \mathcal{X}_1^T r_i(t))^T \hat{\Delta}_i^T \hat{\sigma}_i' \\ \dot{\hat{\theta}}_i &= \mathcal{P}_{3i} \|\mathcal{B}^T \mathcal{X}_1^T r_i(t)\| x_{di}^F \quad \theta_i(0) \geq 0 \end{aligned}$$

where $\mathcal{P}_{1i} > 0$, $\mathcal{P}_{2i} > 0$, $\mathcal{P}_{3i} > 0$, $\rho_i > 0$ for $i \in \mathbb{F}$ are the parameters to be designed, and the matrix $\mathcal{X}_1$ will be given in the following theorem. The $\text{sgn}_p(\gamma) : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is $\text{sgn}_p(\gamma) = \{\text{sgn}(r_1(\gamma)); \text{sgn}(r_2(\gamma)); \dots; \text{sgn}(r_p(\gamma))\}$ and $r_i(\gamma)$ denote the i th rank of γ .

The control flowchart of this article is illustrated in Fig. 1. First, the integral sliding mode surface $s_i(t)$ is constructed based on the states of singular multiagent systems (1) and observer systems (7). Subsequently, the neural-adaptive FTCT protocol is designed to estimate the actuator attacks and drive the states of agents to achieve consensus tracking within a finite time.

Theorem 1: The sliding mode surface (10) can be reached in a finite time T_r with $T_r \leq T$ when the parameters $\rho_i, i \in \mathbb{F}$ in distributed neural-adaptive protocol (11) satisfies

$$\rho_i \geq \frac{\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]}{T} \| (I_N \otimes \mathcal{B}^T \mathcal{X} E) \tilde{r}(0) \|. \quad (12)$$

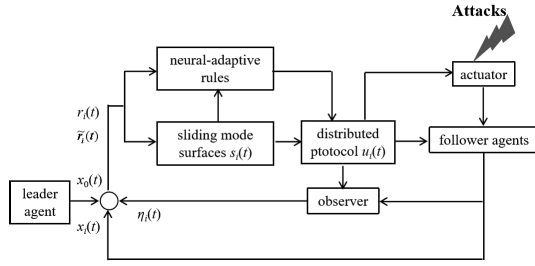


Fig. 1. Control flowchart of this article.

Proof: Define $u(t) = \text{col}\{u_1(t), u_2(t), \dots, u_N(t)\}$, $u_f(t) = \text{col}\{u_{f1}(t), \dots, u_{fN}(t)\}$, $\rho = \text{diag}\{I_p \otimes \rho_1, \dots, I_p \otimes \rho_N\}$ and $\text{sgn}_{Np}(s(t)) = \text{col}\{\text{sgn}_p(s_1(t)), \dots, \text{sgn}_p(s_N(t))\}$. By taking the derivative of sliding surface function $s_i(t)$, one can get that

$$\begin{aligned} \dot{s}(t) &= (\mathcal{L} + \mathcal{J}) \otimes \mathcal{B}^T \mathcal{X} \mathcal{B} \mathcal{K} \tilde{r}(t) \\ &\quad + (I_N \otimes \mathcal{B}^T \mathcal{X} \mathcal{B}) (u(t) + u_f(t)) \\ &= - (I_N \otimes \mathcal{B}^T \mathcal{X} \mathcal{B}) \rho \text{sgn}_{Np}(s(t)). \end{aligned}$$

First, construct the following Lyapunov function:

$$V_s(t) = \frac{1}{2} s^T(t) (I_N \otimes \mathcal{B}^T \mathcal{X} \mathcal{B})^{-1} s(t)$$

where $\mathcal{X} > 0$ is chosen such that $\mathcal{B}^T \mathcal{X} \mathcal{B}$ is invertible.

Then, taking the derivative of the Lyapunov function constructed above along $\dot{s}(t)$ yields

$$\begin{aligned} \dot{V}_s(t) &= s^T(t) (I_N \otimes \mathcal{B}^T \mathcal{X} \mathcal{B})^{-1} \dot{s}(t) \\ &= -s^T(t) \rho \text{sgn}_{Np}(s(t)) \\ &\leq -\rho_{\min} \|s(t)\| \\ &\leq -\frac{\rho_{\min}}{\sqrt{\frac{\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]}{2}}} \sqrt{V_s(t)} \end{aligned}$$

where $\rho_{\min} = \min\{\rho_1, \rho_2, \dots, \rho_N\}$. Then, the setting time T_r has the form $T_r \leq 2(V_s(0))^{1/2}(((\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]/2))^{1/2}/\rho(t))$. According to the constraint (12) of ρ_i , it can be obtained that

$$\begin{aligned} T_r &\leq 2 \sqrt{\frac{\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]}{2}} \frac{\sqrt{\frac{\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]}{2}}}{\rho_{\min}} \|s(0)\| \\ &\leq \frac{\lambda_{\max}[(\mathcal{B}^T \mathcal{X} \mathcal{B})^{-1}]}{\rho_{\min}} \|I_N \otimes \mathcal{B}^T \mathcal{X} E \tilde{r}(0)\| \\ &\leq T. \end{aligned}$$

The above inequality reveals that the reaching time T_r satisfies $T_r \leq T$. The proof is completed. ■

B. FTCT Over Reaching Phase With $[0, T_r]$

Here, the solvability of the FTCT problem for singular multiagent systems (1) under actuator attack is provided during the reaching phase with $[0, T_r]$.

The definitions of symbols are as follows:

$$\begin{aligned} \tilde{\Delta} &= \text{diag}\{\tilde{\Delta}_1, \tilde{\Delta}_2, \dots, \tilde{\Delta}_N\}, \quad \hat{\Delta} = \text{diag}\{\hat{\Delta}_1, \hat{\Delta}_2, \dots, \hat{\Delta}_N\} \\ \tilde{\Pi} &= \text{diag}\{\tilde{\Pi}_1, \tilde{\Pi}_2, \dots, \tilde{\Pi}_N\}, \quad \hat{\Pi} = \text{diag}\{\hat{\Pi}_1, \hat{\Pi}_2, \dots, \hat{\Pi}_N\} \end{aligned}$$

$$\begin{aligned} \tilde{\theta} &= \text{col}\{\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_N\}, \quad \hat{\theta} = \text{col}\{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_N\} \\ w &= \text{col}\{w_1, w_2, \dots, w_N\}, \quad \rho = \text{diag}\{\rho_1, \rho_2, \dots, \rho_N\} \\ \hat{\sigma} &= \text{col}\{\hat{\sigma}_1, \hat{\sigma}_2, \dots, \hat{\sigma}_N\}, \quad \hat{\sigma}' = \text{col}\{\hat{\sigma}'_1, \hat{\sigma}'_2, \dots, \hat{\sigma}'_N\} \\ r(\cdot) &= \text{col}\{r_1(\cdot), r_2(\cdot), \dots, r_N(\cdot)\} \\ \tilde{r}(\cdot) &= \text{col}\{\tilde{r}_1(\cdot), \tilde{r}_2(\cdot), \dots, \tilde{r}_N(\cdot)\} \\ x_d(t) &= \text{col}\{x_{d1}(t), x_{d2}(t), \dots, x_{dN}(t)\} \\ x_d^F(t) &= \text{col}\{x_{d1}^F(t), x_{d2}^F(t), \dots, x_{dN}^F(t)\} \\ \delta &= \text{col}\{\delta(\tilde{x}_{d1}(t)), \delta(\tilde{x}_{d2}(t)), \dots, \delta(\tilde{x}_{dN}(t))\} \\ \text{sgn}_{Np}(\mathcal{B}^T \mathcal{X}_1^T r(t)) &= \text{col}\{\text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r_1(t)) \\ &\quad \text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r_2(t)), \dots, \text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r_N(t))\}. \end{aligned}$$

Based on the proposed distributed neural-adaptive protocol (11), the coupled system can be obtained as

$$\begin{aligned} (I_N \otimes E) \dot{r}(t) &= (I_N \otimes \mathcal{A}_1) r(t) - ((\mathcal{L} + \mathcal{J}) \otimes \mathcal{B} \mathcal{K}) \tilde{r}(t) \\ &\quad + (I_N \otimes \mathcal{A}_2) r(t - d(t)) + (I_N \otimes \mathcal{D}) w(t) \\ &\quad - (I_N \otimes \mathcal{B}) [\tilde{\Delta}^T (\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d) + \tilde{\Delta}^T \hat{\sigma}' \hat{\Pi}^T x_d \\ &\quad + \delta + \hat{\theta}^T x_d^F \text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r(t)) \\ &\quad + \rho \text{sgn}_{Np}(s(t))]. \end{aligned} \quad (13)$$

From (9) and (11), the equation of state at reaching phase with $[0, T_r]$ is calculated as follows:

$$\begin{aligned} (I_N \otimes E) \dot{\tilde{r}}(t) &= (I_N \otimes \mathcal{A}_1 - (\mathcal{L} + \mathcal{J}) \otimes \mathcal{B} \mathcal{K}) \tilde{r}(t) \\ &\quad + (I_N \otimes \mathcal{A}_2) \tilde{r}(t - d(t)) \\ &\quad - (I_N \otimes \mathcal{B}) \rho \text{sgn}_{Np}(s(t)). \end{aligned} \quad (14)$$

Remark 4: Without loss of generality, given the scalars $\tilde{l}_2 > \tilde{l}_1^* > \tilde{l}_1 > 0$ and initial condition $\tilde{r}^T(0)(I_N \otimes E^T \mathcal{R} E) \tilde{r}(0) \leq \tilde{l}_1$. In this work, FTS with respect to $(\tilde{l}_1, \tilde{l}_2, [0, T])$ of error system (9) is also provided.

Theorem 2: For a given reaching phase $[0, T_r]$, the singular multiagent systems (1) are FTCT with respect to $(\tilde{l}_1, \tilde{l}_1^*, [0, T_r])$ and the error systems (9) are FTS with respect to $(\tilde{l}_1, \tilde{l}_1^*, [0, T_r])$, if there exist nonsingular matrices $\mathcal{X}_1, \mathcal{X}_2$, symmetric positive matrices $\mathcal{M}_1 > 0$ and $\mathcal{M}_2 > 0$, matrix $\mathcal{K} \in \mathbb{R}^{q \times n}$ and positive scalars $\tilde{l}_1^*, \tilde{l}_1^*, \alpha$ satisfies

$$\begin{aligned} \mathcal{X}_1 E &= E^T \mathcal{X}_1^T \geq 0 \\ \mathcal{X}_2 E &= E^T \mathcal{X}_2^T \geq 0 \\ \Omega_{6 \times 6} &= \Omega_{6 \times 6}^T < 0 \\ e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\hat{\mathcal{X}}_1)} &< \tilde{l}_1^* \\ e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\hat{\mathcal{X}}_2)} &< \tilde{l}_1^* \\ \mathcal{X}_1 E &= E^T \mathcal{X}_1^T = E^T \mathcal{R}^{\frac{1}{2}} \hat{\mathcal{X}}_1 \mathcal{R}^{\frac{1}{2}} E \\ \mathcal{X}_2 E &= E^T \mathcal{X}_2^T = E^T \mathcal{R}^{\frac{1}{2}} \hat{\mathcal{X}}_2 \mathcal{R}^{\frac{1}{2}} E \\ \mathcal{M}_1 &= \|E^T\| \mathcal{R}^{\frac{1}{2}} \hat{\mathcal{M}}_1 \mathcal{R}^{\frac{1}{2}} \|E\| \\ \mathcal{M}_2 &= \|E^T\| \mathcal{R}^{\frac{1}{2}} \hat{\mathcal{M}}_2 \mathcal{R}^{\frac{1}{2}} \|E\| \end{aligned} \quad (15)$$

where $\Omega_{11} = I_N \otimes (\mathcal{X}_1 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_1^T + \mathcal{M}_1) - \alpha(I_N \otimes \mathcal{X}_1 E)$,

$$\begin{aligned} \Omega_{12} &= I_N \otimes \mathcal{X}_1 \mathcal{A}_2, \quad \Omega_{13} = -(\mathcal{L} + \mathcal{J}) \otimes \mathcal{X}_1 \mathcal{B} \mathcal{K}, \\ \Omega_{15} &= I_N \otimes \mathcal{X}_1 \mathcal{D}, \quad \Omega_{16} = -I_N \otimes \mathcal{X}_1 \mathcal{B}, \end{aligned}$$

$$\begin{aligned}
\Omega_{22} &= (\varepsilon - 1)I_N \otimes \mathcal{M}_1, \\
\Omega_{33} &= I_N \otimes (\mathcal{X}_2 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_2^T + \mathcal{M}_2) - (\mathcal{L} + \mathcal{J}) \otimes (\mathcal{X}_2 \mathcal{B} \mathcal{K} \\
&\quad + \mathcal{K}^T \mathcal{B}^T \mathcal{X}_2^T) - \alpha(I_N \otimes \mathcal{X}_2 E), \quad \Omega_{34} = I_N \otimes \mathcal{X}_2 \mathcal{A}_2, \\
\Omega_{36} &= -I_N \otimes \mathcal{X}_2 \mathcal{B}, \quad \Omega_{44} = -(1 - \varepsilon)I_N \otimes \mathcal{M}_2, \\
\Omega_{55} &= -\alpha I, \quad \Omega_{66} = -\alpha I, \quad \varrho = \max\{\lambda(\hat{\mathcal{X}}_1)\} + d\lambda(\hat{\mathcal{M}}_1), \\
\tilde{\varrho} &= \max\{\lambda(\hat{\mathcal{X}}_2)\} + d\lambda(\hat{\mathcal{M}}_2), \mathcal{P}_1 = \text{diag}\{\mathcal{P}_{11}, \mathcal{P}_{12}, \dots, \mathcal{P}_{1N}\}, \\
\rho_{\max} &= \max\{\rho_1, \rho_2, \dots, \rho_N\}, \mathcal{P}_2 = \text{diag}\{\mathcal{P}_{21}, \mathcal{P}_{22}, \dots, \mathcal{P}_{2N}\}, \\
\mathcal{P}_3 &= \text{diag}\{\mathcal{P}_{31}, \mathcal{P}_{32}, \dots, \mathcal{P}_{3N}\}, \\
\Gamma &= \max\{\lambda(\mathcal{P}_1^{-1})\} \text{tr}(\tilde{\Delta}^T(0)\tilde{\Delta}(0)) + \max\{\lambda(\mathcal{P}_2^{-1})\} \\
&\quad \times \text{tr}(\tilde{\Pi}^T(0)\tilde{\Pi}(0)) + \max\{\lambda(\mathcal{P}_3^{-1})\}(\tilde{\theta}^T(0)\tilde{\theta}(0)).
\end{aligned}$$

Proof: First, we will prove that the pair $(E, \mathcal{A}_1 + \mathcal{A}_2)$ is impulse free. From (15), it is easy to derived that $\tilde{\Omega} = \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ * & \Omega_{22} \end{bmatrix} < 0$. Then, premultiplying and postmultiplying $\tilde{\Omega}$ by $\begin{bmatrix} I & I \\ 0 & I \end{bmatrix}$ and $\begin{bmatrix} I & 0 \\ I & I \end{bmatrix}$, we have $\Omega_{11} + \Omega_{12} + \Omega_{21} + \Omega_{22} < 0$.

For $\text{rank}(E) = r < n$, nonsingular matrices $\mathcal{V}$ and $\mathcal{N}$ can be found and satisfy $\mathcal{V}E\mathcal{N} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$. Without loss of generality, one sets

$$\begin{aligned}
\mathcal{V}\mathcal{A}_1\mathcal{N} &= \begin{bmatrix} \mathcal{A}_{11} & \mathcal{A}_{12} \\ \mathcal{A}_{13} & \mathcal{A}_{14} \end{bmatrix} & \mathcal{V}\mathcal{A}_2\mathcal{N} &= \begin{bmatrix} \mathcal{A}_{21} & \mathcal{A}_{22} \\ \mathcal{A}_{23} & \mathcal{A}_{24} \end{bmatrix} \\
\mathcal{N}^T\mathcal{X}_1\mathcal{V}^{-1} &= \begin{bmatrix} \mathcal{X}_{11} & \mathcal{X}_{12} \\ \mathcal{X}_{13} & \mathcal{X}_{14} \end{bmatrix} & \mathcal{N}^T\mathcal{M}_1\mathcal{N} &= \begin{bmatrix} \mathcal{M}_{11} & \mathcal{M}_{12} \\ \mathcal{M}_{13} & \mathcal{M}_{14} \end{bmatrix}
\end{aligned}$$

where $\mathcal{A}_{11}, \mathcal{A}_{21}, \mathcal{X}_{11}, \mathcal{M}_{11} \in \mathbb{R}^{r \times r}$, $\mathcal{A}_{12}, \mathcal{A}_{22}, \mathcal{X}_{12}, \mathcal{M}_{12} \in \mathbb{R}^{r \times (n-r)}$, $\mathcal{A}_{13}, \mathcal{A}_{23}, \mathcal{X}_{13}, \mathcal{M}_{13} \in \mathbb{R}^{(n-r) \times r}$, and $\mathcal{A}_{14}, \mathcal{A}_{24}, \mathcal{X}_{14}, \mathcal{M}_{14} \in \mathbb{R}^{(n-r) \times (n-r)}$.

For nonsingular matrix $\mathcal{X}_1$, according to constraint $\mathcal{X}_1 E = E^T \mathcal{X}_1^T \geq 0$, it can be obtained that $\mathcal{X}_{11} = \mathcal{X}_{11}^T$, $\mathcal{X}_{13} = 0$, and $\det(\mathcal{X}_{14}) \neq 0$. The matrix $\det(\mathcal{M}_{14}) > 0$ can be deduced from the positive definition of $\mathcal{M}_1$.

Premultiplying and postmultiplying $\Omega_{11} + \Omega_{12} + \Omega_{21} + \Omega_{22}$ by $\mathcal{N}^T$ and $\mathcal{N}$, we have

$$\begin{bmatrix} * & * \\ * & I_N \otimes (\mathcal{X}_{14}(\mathcal{A}_{14} + \mathcal{A}_{24}) + (\mathcal{A}_{14} + \mathcal{A}_{24})^T \mathcal{X}_{14}^T + \varepsilon \mathcal{M}_{14}) \end{bmatrix} < 0$$

where $*$ is unrelated to the proof. It is obviously that $\det(\mathcal{A}_{14} + \mathcal{A}_{24}) \neq 0$. Therefore, the pair $(E, \mathcal{A}_1 + \mathcal{A}_2)$ is impulse free.

Second, we give the following Lyapunov function:

$$V(t) = V_1(t) + V_2(t) + V_3(t) \quad (16)$$

where

$$\begin{aligned}
V_1(t) &= r^T(t)(I_N \otimes \mathcal{X}_1 E) r(t) \\
&\quad + \int_{t-d(t)}^t r^T(\tau)(I_N \otimes \mathcal{M}_1) r(\tau) d\tau \\
V_2(t) &= \tilde{r}^T(t)(I_N \otimes \mathcal{X}_2 E) \tilde{r}(t) \\
&\quad + \int_{t-d(t)}^t \tilde{r}^T(\tau)(I_N \otimes \mathcal{M}_2) \tilde{r}(\tau) d\tau \\
V_3(t) &= \text{tr}(\tilde{\Delta}^T \mathcal{P}_1^{-1} \tilde{\Delta}) + \text{tr}(\tilde{\Pi}^T \mathcal{P}_2^{-1} \tilde{\Pi}) + \tilde{\theta}^T \mathcal{P}_3^{-1} \tilde{\theta}.
\end{aligned}$$

From (2) and (13), taking the derivative of $V_1(t)$ yields

$$\dot{V}_1(t) \leq r^T(t)[I_N \otimes (\mathcal{X}_1 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_1^T + \mathcal{M}_1)] r(t)$$

$$\begin{aligned}
&+ \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{A}_2) r(t - d(t))\} \\
&- (1 - \varepsilon) r^T(t - d(t))(I_N \otimes \mathcal{M}_1) r(t - d(t)) \\
&- \text{sym}\{r^T(t)[(\mathcal{L} + \mathcal{J}) \otimes \mathcal{X}_1 \mathcal{B} \mathcal{K}] \tilde{r}(t)\} \\
&- \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{D}) w(t)\} \\
&- \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{B})[\tilde{\Delta}^T(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d) \\
&+ \hat{\Delta}^T \hat{\sigma}' \tilde{\Pi}^T x_d + \delta + \hat{\theta}^T x_d^F \text{sgn}_{N_p}((I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)) \\
&+ \rho \text{sgn}_{N_p}(s(t))]\}. \quad (17)
\end{aligned}$$

From (2) and error systems (14), taking the derivative of $V_2(t)$ during reaching phase $[0, T_r]$ yields

$$\begin{aligned}
\dot{V}_2(t) &\leq \tilde{r}^T(t)[I_N \otimes (\mathcal{X}_2 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_2^T + \mathcal{M}_2) \\
&- (\mathcal{L} + \mathcal{J}) \otimes (\mathcal{X}_2 \mathcal{B} \mathcal{K} + \mathcal{K}^T \mathcal{B}^T \mathcal{X}_2^T)] \tilde{r}(t) \\
&+ \text{sym}\{\tilde{r}^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{A}_2) \tilde{r}(t - d(t))\} \\
&- (1 - \varepsilon) \tilde{r}^T(t - d(t))(I_N \otimes \mathcal{M}_2) \tilde{r}(t - d(t)) \\
&- \text{sym}\{r^T(t)[I_N \otimes (\mathcal{X}_2 \mathcal{B})] \rho \text{sgn}_{N_p}(s(t))\}. \quad (18)
\end{aligned}$$

The derivative of $V_3(t)$ along with the neural-adaptive rules in (11) is given as

$$\begin{aligned}
\dot{V}_3(t) &= 2\text{tr}(\tilde{\Delta}(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d)(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)) \\
&+ 2\text{tr}(\tilde{\Pi}^T x_d((I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t))^T \hat{\Delta}^T \hat{\sigma}') \\
&+ 2\tilde{\theta}^T \|(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)\| x_d^F. \quad (19)
\end{aligned}$$

Combining some terms in (17) with (19) yield the following inequalities:

$$\begin{aligned}
&- \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{B}) \tilde{\Delta}^T(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d)\} \\
&+ 2\text{tr}(\tilde{\Delta}(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d)(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)) \\
&\leq -2\text{tr}(\tilde{\Delta}(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d)(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)) \\
&+ 2\text{tr}(\tilde{\Delta}(\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d)(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)) \\
&= 0
\end{aligned}$$

and

$$\begin{aligned}
&- \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{B}) \hat{\Delta}^T \hat{\sigma}' \tilde{\Pi}^T x_d\} \\
&+ 2\text{tr}(\tilde{\Pi}^T x_d(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t))^T \hat{\Delta}^T \hat{\sigma}') \\
&\leq 0.
\end{aligned}$$

In addition, by combining $\tilde{\theta} = \hat{\theta} - \theta$ it can be calculated that

$$\begin{aligned}
&2\tilde{\theta}^T \|(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)\| x_d^F - \text{sym}\{r^T(t)(I_N \otimes \mathcal{X}_1 \mathcal{B}) \\
&\times (\delta + \hat{\theta}^T x_d^F \text{sgn}_{N_p}((I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)))\} \\
&\leq 2\tilde{\theta}^T \|(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)\| x_d^F \\
&\quad + 2\|(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)\| \theta^T x_d^F \\
&\quad - 2\|(I_N \otimes \mathcal{B}^T \mathcal{X}_1^T) r(t)\| \hat{\theta}^T x_d^F \\
&= 0.
\end{aligned}$$

Construct the following auxiliary function:

$$\begin{aligned}
Y_1(t) &= \dot{V}(t) - \alpha(\rho \text{sgn}_{N_p}(s(t)))^T (\rho \text{sgn}_{N_p}(s(t))) \\
&\quad - \alpha w^T(t) w(t) - \alpha[r^T(t)(I_N \otimes \mathcal{X}_1 E) r(t) \\
&\quad + \tilde{r}^T(t)(I_N \otimes \mathcal{X}_2 E) \tilde{r}(t)].
\end{aligned}$$

According to (17)–(19) and the definition of Ω , one can get that $Y_1(t) = \xi^T(t) \Omega \xi(t) < 0$ for $\xi(t) = \text{col}\{r(t), r(t - d(t)), \tilde{r}(t), \tilde{r}(t - d(t)), w(t), \rho \text{sgn}_{N_p}(s(t))\}$.

Therefore, the following inequality can be derived:

$$\begin{aligned} \dot{V}(t) &< \alpha [r^T(t)(I_N \otimes \mathcal{X}_1 E) r(t) + \tilde{r}^T(t)(I_N \otimes \mathcal{X}_2 E) \tilde{r}(t)] \\ &\quad + \alpha (\rho \text{sgn}_{N_p}(s(t)))^T (\rho \text{sgn}_{N_p}(s(t))) + \alpha w^T(t) w(t) \\ &\leq \alpha V(t) + \alpha (\rho \text{sgn}_{N_p}(s(t)))^T (\rho \text{sgn}_{N_p}(s(t))) \\ &\quad + \alpha w^T(t) w(t). \end{aligned}$$

Multiplying both sides of the above equation by $e^{-\alpha t}$, and integrating from 0 to T_r yields

$$\begin{aligned} e^{-\alpha t} V(t) &< V(0) + \alpha \int_0^{T_r} e^{-\alpha \tau} (\rho(\tau) \text{sgn}_{N_p}(s(\tau)))^T \\ &\quad \times (\rho(\tau) \text{sgn}_{N_p}(s(\tau))) d\tau + \alpha \int_0^{T_r} e^{-\alpha \tau} w^T(\tau) w(\tau) d\tau \\ &< V(0) + \alpha \rho_{\max} N T + \alpha N \varpi. \end{aligned}$$

Based on the definition of ϱ , $\tilde{\varrho}$ and $\Gamma(0)$, it follows from (16) and Lemma 4 that:

$$\begin{aligned} V(0) &\leq r^T(0) [I_N \otimes \mathcal{X}_1 E + d(I_N \otimes \mathcal{M}_1)] r(0) \\ &\quad + \tilde{r}^T(0) [I_N \otimes \mathcal{X}_2 E + d(I_N \otimes \mathcal{M}_2)] \tilde{r}(0) \\ &\quad + \lambda_{\max}(\mathcal{P}_1^{-1}) \text{tr}(\tilde{\Delta}^T(0) \tilde{\Delta}(0)) + \lambda_{\max}(\mathcal{P}_2^{-1}) \\ &\quad \times \text{tr}(\tilde{\Pi}^T(0) \tilde{\Pi}(0)) + \lambda_{\max}(\mathcal{P}_3^{-1}) \tilde{\theta}^T(0) \tilde{\theta}(0) \\ &\leq \varrho l_1 + \tilde{\varrho} \tilde{l}_1 + \Gamma. \end{aligned}$$

On the other hand, from (16), one has

$$e^{-\alpha t} V(t) \geq e^{-\alpha T} \lambda_{\min}(\hat{\mathcal{X}}_1) r^T(t) (I_N \otimes E^T \mathcal{R} E) r(t).$$

And we can derive the following result:

$$\begin{aligned} r^T(t) (I_N \otimes E^T \mathcal{R} E) r(t) \\ \leq e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\hat{\mathcal{X}}_1)} < l_1^*. \end{aligned}$$

From (16), the $V(t)$ function and $\tilde{r}(t)$ satisfy that

$$e^{-\alpha t} V(t) \geq e^{-\alpha T} \lambda_{\min}(\hat{\mathcal{X}}_2) \tilde{r}^T(t) (I_N \otimes E^T \mathcal{R} E) \tilde{r}(t).$$

Then, the following FTS results can be deduced:

$$\begin{aligned} \tilde{r}^T(t) (I_N \otimes E^T \mathcal{R} E) \tilde{r}(t) \\ \leq e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\hat{\mathcal{X}}_2)} < \tilde{l}_1^*. \end{aligned}$$

Therefore, the singular multiagent systems (1) are FTCT with respect to $(l_1, l_1^*, [0, T_r])$ and the error systems (9) are FTS with respect to $(\tilde{l}_1, \tilde{l}_1^*, [0, T_r])$. The proof is completed. ■

Remark 5: For the FTCT proof of the sliding motion phase within $[T_r, T]$, the following estimation results need to be provided. From (16) and Lemma 4, we can infer that $e^{-\alpha t} V(t) \geq e^{-\alpha T} \lambda_{\min}(\mathcal{P}_1^{-1}) \text{tr}(\tilde{\Delta}^T \tilde{\Delta})$. And $\text{tr}(\tilde{\Delta}^T \tilde{\Delta})$ satisfies the following constraints:

$$\text{tr}(\tilde{\Delta}^T \tilde{\Delta}) \leq e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\mathcal{P}_1^{-1})} < \nu_1.$$

Similar to the above proof, it can be obtained that

$$\text{tr}(\tilde{\Pi}^T \tilde{\Pi}) \leq e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\mathcal{P}_2^{-1})} < \nu_2$$

and

$$\tilde{\theta}^T \tilde{\theta} \leq e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T}{\lambda_{\min}(\mathcal{P}_3^{-1})} < \nu_3.$$

C. FTCT Over Sliding Motion Phase Within $[T_r, T]$

In this subsection, the solvability conditions for the FTCT problem for singular multiagent systems in the interval $[T_r, T]$ (i.e., sliding motion phase) are analyzed.

In the sliding motion phase within $[T_r, T]$, $s_i(t) = 0$ and $\dot{s}_i(t) = 0$ are true. According to $\dot{s}_i(t) = 0$, the equivalent controller $u_{\text{eqi}}(t)$ has the following form:

$$u_{\text{eqi}}(t) = \mathcal{K} \left[\sum_{j \in \mathcal{N}_i} a_{ij} (\tilde{r}_j(t) - \tilde{r}_i(t)) - J_i \tilde{r}_i(t) \right] - u_{fi}(t). \quad (20)$$

Therefore, the coupled systems in $[T_r, T]$ are given as

$$\begin{aligned} (I_N \otimes E) \dot{r}(t) &= (I_N \otimes \mathcal{A}_1) r(t) - ((\mathcal{L} + \mathcal{J}) \otimes \mathcal{B} \mathcal{K}) \tilde{r}(t) \\ &\quad + (I_N \otimes \mathcal{A}_2) r(t - d(t)) - (I_N \otimes \mathcal{B}) \\ &\quad \times [\tilde{\Delta}^T (\hat{\sigma} - \hat{\sigma}' \hat{\Pi}^T x_d) + \tilde{\Delta}^T \hat{\sigma}' \hat{\Pi}^T x_d + \delta \\ &\quad + \hat{\theta}^T x_d^F \text{sgn}_p(\mathcal{B}^T \mathcal{X}_1^T r(t))] + (I_N \otimes \mathcal{D}) w(t) \end{aligned} \quad (21)$$

and

$$\begin{aligned} (I_N \otimes E) \dot{\tilde{r}}(t) &= (I_N \otimes \mathcal{A}_1 - (\mathcal{L} + \mathcal{J}) \otimes \mathcal{B} \mathcal{K}) \tilde{r}(t) \\ &\quad + (I_N \otimes \mathcal{A}_2) \tilde{r}(t - d(t)). \end{aligned} \quad (22)$$

Theorem 3: The singular multiagent systems (1) are FTCT respect to $(l_1^*, l_2, [T_r, T])$ and the error systems (9) are FTS respect to $(\tilde{l}_1^*, \tilde{l}_2, [T_r, T])$, if there exist nonsingular matrices $\mathcal{X}_1, \mathcal{X}_2$, symmetric positive matrices $\mathcal{M}_1 > 0$ and $\mathcal{M}_2 > 0$, matrix $\mathcal{K} \in \mathbb{R}^{q \times n}$ and positive scalars $l_1^*, \tilde{l}_1^*, \alpha$ satisfies

$$\begin{aligned} \mathcal{X}_1 E &= E^T \mathcal{X}_1^T \geq 0 \\ \mathcal{X}_2 E &= E^T \mathcal{X}_2^T \geq 0 \\ \hat{\Omega} &= \begin{bmatrix} \Omega_{11} & \Omega_{12} & \Omega_{13} & 0 & I_N \otimes \mathcal{X}_1 \mathcal{D} \\ * & \Omega_{22} & 0 & 0 & 0 \\ * & * & \Omega_{33} & I_N \otimes \mathcal{X}_2 \mathcal{A}_2 & 0 \\ * & * & * & -(1 - \varepsilon) I_N \otimes \mathcal{M}_2 & 0 \\ * & * & * & * & -\alpha I \end{bmatrix} < 0 \\ e^{\alpha T} \frac{l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W}{\lambda_{\min}(\hat{\mathcal{X}}_1)} &< l_2 \\ e^{\alpha T} \frac{l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W}{\lambda_{\min}(\hat{\mathcal{X}}_2)} &< \tilde{l}_2 \\ l_1 &< l_1^* < l_2 \\ \tilde{l}_1 &< \tilde{l}_1^* < \tilde{l}_2 \end{aligned} \quad (23)$$

where $\Omega_{11}, \Omega_{12}, \Omega_{13}, \Omega_{22}$, and Ω_{33} are the same as Theorem 2, and $W = \lambda_{\max}(\mathcal{P}_1^{-1}) \nu_1 + \lambda_{\max}(\mathcal{P}_2^{-1}) \nu_2 + \lambda_{\max}(\mathcal{P}_3^{-1}) \nu_3$.

Proof: The proof of impulse free for the closed-loop systems is the same as Theorem 2, and it is omitted. The Lyapunov function $V(t) = V_1(t) + V_2(t) + V_3(t)$ is given in (16). The auxiliary function is constructed as

$$\begin{aligned} Y_2(t) &= \dot{V}(t) - \alpha w^T(t) w(t) - \alpha [r^T(t) (I_N \otimes \mathcal{X}_1 E) r(t) \\ &\quad + \tilde{r}^T(t) (I_N \otimes \mathcal{X}_2 E) \tilde{r}(t)]. \end{aligned}$$

Define the vector $\hat{\xi}(t) = \text{col}\{r(t), r(t - d(t)), \tilde{r}(t), \tilde{r}(t - d(t)), w(t)\}$, and take the derivative of the $V(t)$ function along

(21) and (22). Then, $Y_2(t) = \hat{\xi}^T(t)\hat{\Omega}\hat{\xi}(t) < 0$ can be obtained by calculation. This also indicates that

$$\begin{aligned} \dot{V}(t) &< \alpha [r^T(t)(I_N \otimes \mathcal{X}_1 E) r(t) + \tilde{r}^T(t)(I_N \otimes \mathcal{X}_2 E) \tilde{r}(t)] \\ &\quad + \alpha w^T(t) w(t) \\ &\leq \alpha V(t) + \alpha w^T(t) w(t). \end{aligned} \quad (24)$$

Multiply both sides of (24) by $e^{-\alpha t}$ and integrating from T_r to T , it has

$$\begin{aligned} e^{-\alpha T} V(t) &< V(T_r) + \alpha \int_{T_r}^T e^{-\alpha \tau} w^T(\tau) w(\tau) d\tau \\ &< V(T_r) + \alpha N \varpi. \end{aligned} \quad (25)$$

According to Lemma 4, it yields from the definition of $V(t)$ and Remark 5 that

$$\begin{aligned} V(T_r) &< r^T(T_r)[I_N \otimes \mathcal{X}_1 E + d(I_N \otimes \mathcal{M}_1)] r(T_r) \\ &\quad + \tilde{r}^T(T_r)[I_N \otimes \mathcal{X}_2 E + d(I_N \otimes \mathcal{M}_2)] \tilde{r}(T_r) \\ &\quad + \lambda_{\max}(\mathcal{P}_1^{-1}) \text{tr}(\tilde{\Delta}^T \tilde{\Delta}) + \lambda_{\max}(\mathcal{P}_2^{-1}) \text{tr}(\tilde{\Pi}^T \tilde{\Pi}) \\ &\quad + \lambda_{\max}(\mathcal{P}_3^{-1}) \tilde{\theta}^T \tilde{\theta} \\ &\leq \varrho l_1^* + \tilde{\varrho} \tilde{l}_1^* + W. \end{aligned} \quad (26)$$

In addition, one has

$$e^{-\alpha t} V(t) \geq e^{-\alpha T} \lambda_{\min}(\hat{\mathcal{X}}_1) r^T(t)(I_N \otimes E^T \mathcal{R} E) r(t). \quad (27)$$

By combining (24)–(27), it can be obtained that

$$\begin{aligned} r^T(t)(I_N \otimes E^T \mathcal{R} E) r(t) \\ \leq e^{\alpha T} \frac{l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W}{\lambda_{\min}(\hat{\mathcal{X}}_1)} < l_2. \end{aligned}$$

Furthermore, the following result is true:

$$\begin{aligned} \tilde{r}^T(t)(I_N \otimes E^T \mathcal{R} E) \tilde{r}(t) \\ \leq e^{\alpha T} \frac{l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W}{\lambda_{\min}(\hat{\mathcal{X}}_2)} < \tilde{l}_2. \end{aligned}$$

According to Definition 3, the singular multiagent systems (1) are FTCT respect to $(l_1^*, l_2, [T_r, T])$ and the error systems (9) are FTS respect to $(\tilde{l}_1^*, \tilde{l}_2, [T_r, T])$. The proof is completed. ■

Remark 6: The block diagram of the FTCT procedure is given in Fig. 2. In Theorem 1, the finite-time reachability of sliding surface $s_i(t)$ designed in (10) is established, and it is proved that $T_r \leq T$. In reaching phase, the FTCT of systems (1) with respect to $(l_1, l_1^*, [0, T_r])$ and the FTS of systems (9) with respect to $(\tilde{l}_1, \tilde{l}_1^*, [0, T_r])$ under the neural-adaptive protocol (11) are provided in Theorem 2. In sliding motion phase, the singular multiagent systems (1) are FTCT respect to $(l_1^*, l_2, [T_r, T])$ and the error systems (9) are FTS respect to $(\tilde{l}_1^*, \tilde{l}_2, [T_r, T])$ under the equivalent controller $u_{eq}(t)$ in (20) also given in Theorem 3. Therefore, the FTCT of singular multiagent systems (1) in the interval $[0, T]$ is proven. The FTS of error systems (9) over the interval $[0, T]$ is also valid.

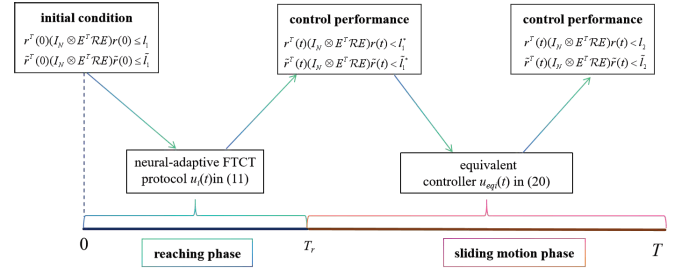


Fig. 2. Block diagram of FTCT procedure.

D. Solving the Gain Matrix $\mathcal{K}$

In this section, the nonlinear inequalities in Theorem 2 and Theorem 3 are linearized. Then, the sufficient conditions for solving the feedback control gain $\mathcal{K}$ in the integral sliding mode surface (10) are presented.

Theorem 4: Given $[0, T]$ and feasible scalar $\alpha > 0$, the singular multiagent systems (1) are FTCT with respect to $(l_1, l_2, [0, T])$ and the error systems (9) are FTS with respect to $(\tilde{l}_1, \tilde{l}_2, [0, T])$, if there exist nonsingular matrices $\mathcal{X}_1, \mathcal{X}_2$, symmetric matrices $\mathcal{M}_1 > 0$ and $\mathcal{M}_2 > 0$, matrix $\hat{\mathcal{K}} \in \mathbb{R}^{q \times n}$ and positive scalars $l_1^*, \tilde{l}_1^*, \varsigma, \kappa_1, \kappa_2, \kappa_3, \kappa_4, \kappa_5$, and κ_6 satisfies

$$\begin{aligned} \mathcal{X}_1 E &= E^T \mathcal{X}_1^T \geq 0 \\ \mathcal{X}_2 E &= E^T \mathcal{X}_2^T \geq 0 \\ \Psi_{8 \times 8} &= \Psi_{8 \times 8}^T < 0 \\ l_1(\kappa_2 + d\kappa_5) + \tilde{l}_1(\kappa_4 + d\kappa_6) + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T \\ &\quad - e^{-\alpha T} l_1^* \kappa_1 < 0 \\ l_1(\kappa_2 + d\kappa_5) + \tilde{l}_1(\kappa_4 + d\kappa_6) + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 N T \\ &\quad - e^{-\alpha T} \tilde{l}_1^* \kappa_3 < 0 \\ l_1^*(\kappa_2 + d\kappa_5) + \tilde{l}_1^*(\kappa_4 + d\kappa_6) + \alpha N \varpi + W - e^{-\alpha T} l_2 \kappa_1 < 0 \\ l_1^*(\kappa_2 + d\kappa_5) + \tilde{l}_1^*(\kappa_4 + d\kappa_6) + \alpha N \varpi + W - e^{-\alpha T} \tilde{l}_2 \kappa_3 < 0 \\ l_1 &< l_1^* < l_2 \\ \tilde{l}_1 &< \tilde{l}_1^* < \tilde{l}_2 \end{aligned} \quad (28)$$

where

$$\begin{aligned} \Psi_{11} &= \mathcal{X}_1 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_1^T + \mathcal{M}_1 - \alpha \mathcal{X}_1 E, \quad \Psi_{12} = \mathcal{X}_1 \mathcal{A}_2 \\ \Psi_{15} &= \mathcal{X}_1 \mathcal{D}, \quad \Psi_{16} = -\mathcal{X}_1 \mathcal{B}, \quad \Psi_{17} = -\lambda_i \mathcal{X}_1 \mathcal{B} \\ \Psi_{22} &= -(1 - \varepsilon) \mathcal{M}_1, \quad \Psi_{33} = \mathcal{X}_2 \mathcal{A}_1 + \mathcal{A}_1^T \mathcal{X}_2^T + \mathcal{M}_2 - \alpha \mathcal{X}_2 E \\ \Psi_{34} &= \mathcal{X}_2 \mathcal{A}_2, \quad \Psi_{36} = -\mathcal{X}_2 \mathcal{B}, \quad \Psi_{37} = -\lambda_i \mathcal{X}_2 \mathcal{B}, \quad \Psi_{38} = \hat{\mathcal{K}}^T \\ \Psi_{44} &= -(1 - \varepsilon) \mathcal{M}_2, \quad \Psi_{55} = -\alpha I, \quad \Omega_{66} = -\alpha I, \\ \Psi_{77} &= -\varsigma I, \quad \Psi_{88} = -\varsigma I, \quad \hat{\mathcal{X}}_1 = \mathcal{R}^{-\frac{1}{2}} \mathcal{S}_1 \mathcal{R}^{-\frac{1}{2}} \\ \hat{\mathcal{X}}_2 &= \mathcal{R}^{-\frac{1}{2}} \mathcal{S}_2 \mathcal{R}^{-\frac{1}{2}}, \quad \hat{\mathcal{M}}_1 = \frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_1 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|} \\ \hat{\mathcal{M}}_2 &= \frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_2 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|} \end{aligned}$$

and the feedback matrix $\mathcal{K} = \frac{\hat{\mathcal{K}}^T}{\varsigma}$.

Proof: First, the solvability of $\Omega < 0$ in Theorem 2 and $\hat{\Omega} < 0$ in Theorem 3 are provided. For the symmetric positive

matrix $\mathcal{L} + \mathcal{J}$, an orthogonal matrix U exist for the following equation:

$$U^T (\mathcal{L} + \mathcal{J}) U = \text{diag} \{\lambda_1, \lambda_2, \dots, \lambda_N\}$$

where $\lambda_i > 0$.

Set the first six rows and six columns of matrix Ψ as $\hat{\Psi}$. Premultiplying and postmultiply matrix Ω in Theorem 2 by $\mathcal{T}_l = \text{diag}\{U^T \otimes I, U^T \otimes I, U^T \otimes I, U^T \otimes I, U^T \otimes I, U^T \otimes I\}$ and $\mathcal{T}_r = \text{diag}\{U \otimes I, U \otimes I, U \otimes I, U \otimes I, U \otimes I, U \otimes I\}$, respectively. The following result is derived from (15) and Schur complement theorem:

$$\begin{aligned} & \mathcal{T}_l \Omega \mathcal{T}_r \\ &= \hat{\Psi} + \text{sym} \left(\begin{bmatrix} -\lambda_i \mathcal{X}_1 \mathcal{B} \\ 0 \\ -\lambda_i \mathcal{X}_2 \mathcal{B} \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \mathcal{K}^T \\ 0 \\ 0 \\ 0 \end{bmatrix}^T \right) \\ &\leq \hat{\Psi} + \varsigma \begin{bmatrix} 0 \\ 0 \\ \mathcal{K}^T \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \mathcal{K}^T \\ 0 \\ 0 \\ 0 \end{bmatrix}^T + \frac{1}{\varsigma} \begin{bmatrix} -\lambda_i \mathcal{X}_1 \mathcal{B} \\ 0 \\ -\lambda_i \mathcal{X}_2 \mathcal{B} \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -\lambda_i \mathcal{X}_1 \mathcal{B} \\ 0 \\ -\lambda_i \mathcal{X}_2 \mathcal{B} \\ 0 \\ 0 \\ 0 \end{bmatrix}^T \\ &< 0 \end{aligned}$$

for any positive ς , and $\mathcal{K} = (\hat{\mathcal{K}}/\varsigma)$. If $\Omega < 0$ is established, it is easy to obtain that the $\hat{\Omega} < 0$ in Theorem 3 is solvable. Therefore, the nonlinear matrix inequalities Ω and $\hat{\Omega}$ are transformed into a linear matrix inequality Ψ for solving.

Next, the expressions for matrices $\hat{\mathcal{X}}_1$, $\hat{\mathcal{X}}_2$, $\hat{\mathcal{M}}_1$, and $\hat{\mathcal{M}}_2$ are given. The matrix E satisfies $\text{rank}(E) = r \leq n$, and the matrix $\Lambda \in \mathbb{R}^{n \times (n-r)}$ satisfying $E^T \Lambda = 0$ can be found. Given any matrix $\Phi_1, \Phi_2 \in \mathbb{R}^{n \times (n-r)}$ and symmetric positive definite matrices $\mathcal{S}_1, \mathcal{S}_2$, one has

$$\mathcal{X}_1 = E^T \mathcal{S}_1 + \Phi_1 \Lambda^T, \quad \mathcal{X}_2 = E^T \mathcal{S}_2 + \Phi_2 \Lambda^T. \quad (29)$$

Substituting (29) into (28) yields $\mathcal{X}_1 E = E^T \mathcal{X}_1^T = E^T \mathcal{S}_1 E \geq 0$ and $\mathcal{X}_2 E = E^T \mathcal{X}_2^T = E^T \mathcal{S}_2 E \geq 0$. Then, the matrices $\hat{\mathcal{X}}_1$ and $\hat{\mathcal{X}}_2$ have the following form:

$$\hat{\mathcal{X}}_1 = \mathcal{R}^{-\frac{1}{2}} \mathcal{S}_1 \mathcal{R}^{-\frac{1}{2}}, \quad \hat{\mathcal{X}}_2 = \mathcal{R}^{-\frac{1}{2}} \mathcal{S}_2 \mathcal{R}^{-\frac{1}{2}}. \quad (30)$$

According to Theorem 2, $\mathcal{M}_1 = \|E^T\| \mathcal{R}^{(1/2)} \hat{\mathcal{M}}_1 \mathcal{R}^{(1/2)} \|E\|$ and $\mathcal{M}_2 = \|E^T\| \mathcal{R}^{(1/2)} \hat{\mathcal{M}}_2 \mathcal{R}^{(1/2)} \|E\|$, the matrix $\hat{\mathcal{M}}_1$ and $\hat{\mathcal{M}}_2$ are solved as follows:

$$\hat{\mathcal{M}}_1 = \frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_1 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|}, \quad \hat{\mathcal{M}}_2 = \frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_2 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|}. \quad (31)$$

Based on the above analysis, we can assume that

$$\begin{aligned} \kappa_1 &= \lambda_{\min} \left(\mathcal{R}^{-\frac{1}{2}} \mathcal{S}_1 \mathcal{R}^{-\frac{1}{2}} \right), & \kappa_2 &= \lambda_{\max} \left(\mathcal{R}^{-\frac{1}{2}} \mathcal{S}_1 \mathcal{R}^{-\frac{1}{2}} \right), \\ \kappa_3 &= \lambda_{\min} \left(\mathcal{R}^{-\frac{1}{2}} \mathcal{S}_2 \mathcal{R}^{-\frac{1}{2}} \right), & \kappa_4 &= \lambda_{\max} \left(\mathcal{R}^{-\frac{1}{2}} \mathcal{S}_2 \mathcal{R}^{-\frac{1}{2}} \right), \\ \kappa_5 &= \lambda_{\max} \left(\frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_1 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|} \right), & \kappa_6 &= \lambda_{\max} \left(\frac{\mathcal{R}^{-\frac{1}{2}} \mathcal{M}_2 \mathcal{R}^{-\frac{1}{2}}}{\|E^T\| \|E\|} \right). \end{aligned} \quad (32)$$

From (30)–(32), the matrices $\hat{\mathcal{X}}_1$, $\hat{\mathcal{X}}_2$, $\hat{\mathcal{M}}_1$, and $\hat{\mathcal{M}}_2$ satisfy

$$\begin{aligned} \kappa_1 I &\leq \hat{\mathcal{X}}_1 \leq \kappa_2 I, & \kappa_3 I &\leq \hat{\mathcal{X}}_2 \leq \kappa_4 I \\ \hat{\mathcal{M}}_1 &\leq \kappa_5 I, & \hat{\mathcal{M}}_2 &\leq \kappa_6 I. \end{aligned}$$

The definitions of $\varrho = \max\{\lambda(\hat{\mathcal{X}}_1)\} + d\lambda(\hat{\mathcal{M}}_1)$ and $\tilde{\varrho} = \max\{\lambda(\hat{\mathcal{X}}_2)\} + d\lambda(\hat{\mathcal{M}}_2)$ are given in Theorem 2, and one can infer that

$$\begin{aligned} & l_1 (\kappa_2 + d\kappa_5) + \tilde{l}_1 (\kappa_4 + d\kappa_6) + \Gamma + \alpha N \varpi \\ & + \alpha \rho_{\max}^2 \text{NT} - e^{-\alpha T} l_1^* \kappa_1 < 0 \\ \Rightarrow & e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 \text{NT}}{\lambda_{\min}(\hat{\mathcal{X}}_1)} < l_1^* \\ & l_1 (\kappa_2 + d\kappa_5) + \tilde{l}_1 (\kappa_4 + d\kappa_6) + \Gamma + \alpha N \varpi \\ & + \alpha \rho_{\max}^2 \text{NT} - e^{-\alpha T} \tilde{l}_1^* \kappa_3 < 0 \\ \Rightarrow & e^{\alpha T} \frac{l_1 \varrho + \tilde{l}_1 \tilde{\varrho} + \Gamma + \alpha N \varpi + \alpha \rho_{\max}^2 \text{NT}}{\lambda_{\min}(\hat{\mathcal{X}}_2)} < \tilde{l}_1^*. \end{aligned}$$

Furthermore, the conditions $e^{\alpha T} (l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W/\lambda_{\min}(\hat{\mathcal{X}}_1)) < l_2$ and $e^{\alpha T} (l_1^* \varrho + \tilde{l}_1^* \tilde{\varrho} + \alpha N \varpi + W/\lambda_{\min}(\hat{\mathcal{X}}_2)) < \tilde{l}_2$ in Theorem 2 can be simplified as

$$\begin{aligned} l_1^* (\kappa_2 + d\kappa_5) + \tilde{l}_1^* (\kappa_4 + d\kappa_6) + \alpha N \varpi + W - e^{-\alpha T} l_2 \kappa_1 &< 0 \\ l_1^* (\kappa_2 + d\kappa_5) + \tilde{l}_1^* (\kappa_4 + d\kappa_6) + \alpha N \varpi + W - e^{-\alpha T} \tilde{l}_2 \kappa_3 &< 0. \end{aligned}$$

That means the FTCT conditions for singular multiagent system (1) in Theorem 2 and Theorem 3 are solvable. The proof is completed. ■

Remark 7: According to the Theorem 4, the solutions for gain matrix $\mathcal{K}$ based on the calculation of $(4n + p + 3q) \times (4n + p + 3q)$ -dimensional matrices, maximum and minimum eigenvalues of the Laplacian matrix $\mathcal{L}$. The solutions for upper bounded l_2 and $\tilde{l}_2$ are dependent on the calculation of $np \times np$, $nl_i \times nl_i$, $4n \times 4n$, and $n \times n$ -dimensional matrices. The increase in dimension n , p , and q of multiagent systems leads to higher computational complexity of FTCT. The computation of both the maximum and minimum eigenvalues for the Laplacian matrix $\mathcal{L}$ remains tractable as the number of agents increases. Therefore, the method proposed in this article demonstrates excellent scalability when it comes to the increase in the number of singular multiagent systems.

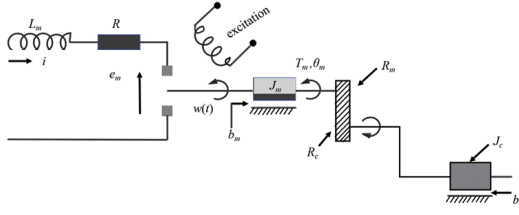
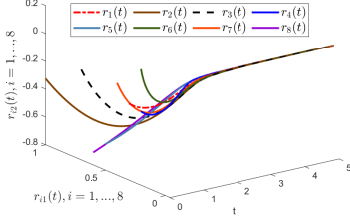
IV. SIMULATION EXAMPLES

The following examples are given to verify the feasibility of FTCT for singular multiagent systems.

Example 1: The block diagram description of i th DC motor is shown in Fig. 3, where $I_i(t)$ is electric current, $\bar{\omega}_i(t)$ represents the speed of the shaft at time t and $u_i(t)$ denotes voltage. According to the basic electrical and mechanical laws, when the inductance L_m of the DC motor is neglected [19], one has

$$\begin{cases} \dot{\bar{\omega}}_i(t) = -\frac{b}{J} \bar{\omega}_i(t) + \frac{K_t}{J} I_i(t) \\ u_i(t) = K_{\bar{\omega}} \bar{\omega}_i(t) + R I_i(t) \end{cases}$$

where $K_{\bar{\omega}}$ is electromotive force constant, K_t is torque constant, and R is electric resistor.

Fig. 3. Block diagram of i th DC motor.Fig. 4. State trajectories of consensus tracking error $r_i(t)$, $i \in \mathbb{F}$.

Let J_m and J_c denote the moments of the motor and the load. b_m and b_c represent the damping ratios with gear ratio m . Define $J = J_m + (J_c/m^2)$, $b = b_m + (b_c/m^2)$, and $x(t) = \text{col}\{\bar{\omega}_i(t), I_i(t)\}$, one has the singular dynamic of i th agent is as follows:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \dot{x}_i(t) = \begin{bmatrix} -\frac{b}{J} & \frac{K_t}{J} \\ K_{\bar{\omega}} & R \end{bmatrix} x_i(t) + \begin{bmatrix} 0 \\ -1 \end{bmatrix} u_i(t).$$

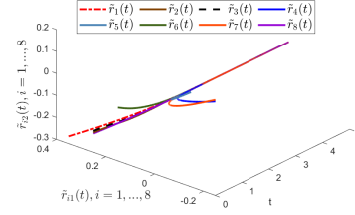
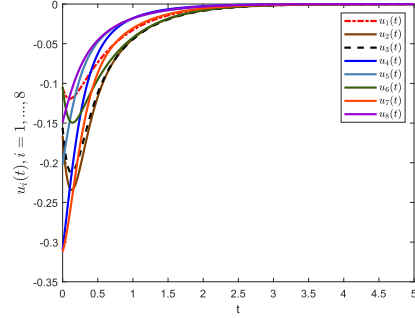
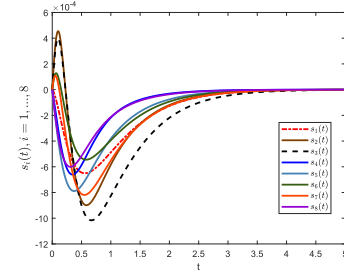
Assume that each DC motor is an agent and the actuator attack is $f_i(t) = x_{i1}(t)\sin(x_{i1}(t))$. The Laplacian matrix of communication topology between eight agents is $\mathcal{L} = \begin{bmatrix} 3 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\ -1 & 3 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 4 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & -1 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix}$ and $\mathcal{J} = \text{diag}\{1, 0, 0, 0, 0, 0, 0, 0\}$. Given $J_m = 0.5\text{kg} \cdot \text{m}$, $J_c = 50\text{kg} \cdot \text{m}$, $b_c = 100$, $R = 1\Omega$, $b_m = 1$, $K_t = 3\text{Nm/A}$, $K_{\bar{\omega}} = 1\text{Vs/rad}$, and $m = 10$.

According to Theorem 4, we have

$$\mathcal{X}_1 = \begin{bmatrix} 0.1792 & -0.1406 \\ 0 & -0.2482 \end{bmatrix}, \quad \mathcal{X}_2 = \begin{bmatrix} 0.1873 & -0.1521 \\ 0 & -0.2567 \end{bmatrix} \\ \mathcal{K} = [-0.1519 \quad -0.0047].$$

For the neural network (4), the number of hidden layer node is 20. The $I_{p,q} \in \mathbb{R}^{p \times q}$ is a matrix with all elements equal to 1, and $\hat{\Delta}_i(0) = 0.01I_{1,20}$, $\hat{\Pi}_i(0) = 0.01I_{3,20}$, $\hat{\theta}_i(0) = 0.01I_{4,1}$ for $i \in \mathbb{F}$. The parameters $\mathcal{P}_{1i} = 5\text{diag}(1, \dots, 1) \in \mathbb{R}^{20 \times 20}$, $\mathcal{P}_{2i} = 5\text{diag}(1, 1, 1)$, and $\mathcal{P}_{3i} = 5\text{diag}(1, 1, 1)$. Set $\mathcal{R} = I$, $x_0(0) = [0.7, -0.7]^T$, $x_1(0) = [0.4, -0.2763]^T$, $x_2(0) = [1, -0.3021]^T$, $x_3(0) = [0.9, -0.3213]^T$, $x_4(0) = [0.5, -0.6697]^T$, $x_5(0) = [0.3, -0.4212]^T$, $x_6(0) = [0.6, -0.2926]^T$, $x_7(0) = [0.8, -0.6417]^T$, and $x_8(0) = [0.2, -0.2686]^T$. Given $T = 5$, the singular multiagent systems are FTCT for $l_1 = 0.69$, $l_1^* = 2.83$, and $l_2 = 94.34$. The error systems $\tilde{r}(t)$ are FTS for $\tilde{l}_1 = 0.48$, $\tilde{l}_1^* = 2.78$, and $\tilde{l}_2 = 91.78$.

From Fig. 4, the state of $r_i(t) = x_i(t) - x_0(t)$, $i = 1, \dots, 8$ is stable, that is, the electric current $I_i(t)$ and the speed of the shaft $\bar{\omega}_i(t)$ of follower agents can be consistent with leader

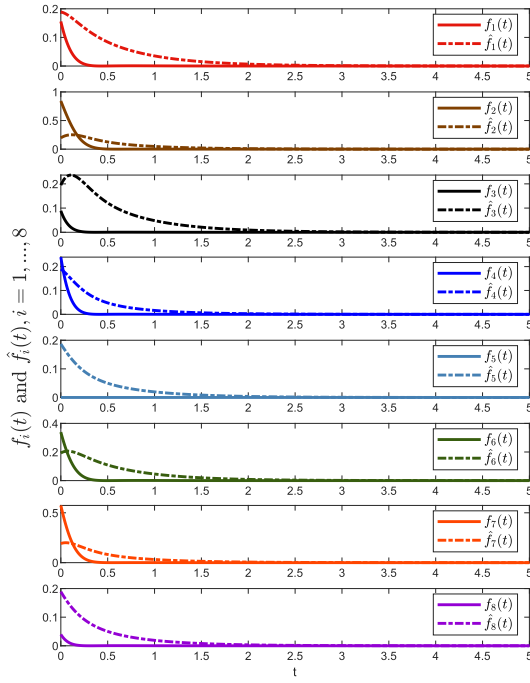
Fig. 5. State trajectories of $\tilde{r}_i(t)$, $i \in \mathbb{F}$.Fig. 6. State trajectories of $u_i(t)$, $i \in \mathbb{F}$.Fig. 7. State trajectories of $s_i(t)$, $i \in \mathbb{F}$.

agent in finite-time. Moreover, the state trajectories of $\tilde{r}_i(t)$, $i \in \mathbb{F}$ within 0–5s are shown in Fig. 5. The simulation results in Fig. 5 demonstrate that $\tilde{r}_i(t)$, $i \in \mathbb{F}$ can reach stability within the specified time.

The state trajectories of neural-adaptive FTCT protocol designed in (11) are shown in Fig. 6. From Fig. 6, the voltage of the DC motor is stable in finite time. Given $\mathcal{X} = I$, the sliding mode function $s_i(t)$, $i \in \mathbb{F}$ can reach stability in about 4s, which has been displayed in Fig. 7. According to Fig. 8, neural network-based estimation function $\hat{f}_i$ can accurately estimate the nonlinear actuator attacks f_i . The accurate estimation time is about 2.5s, which indicates that the estimation method based on the neural network method has a fast estimation speed.

Example 2: For the time-varying delay singular multiagent systems (1), the matrix parameters are given in the following:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathcal{A}_1 = \begin{bmatrix} 0.1 & 2.1 & 1.0 \\ 3.2 & 2.1 & 4.2 \\ 1.0 & 1.9 & 1.0 \end{bmatrix} \\ \mathcal{A}_2 = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 0.01 \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0.1 \\ 0.2 \\ 0 \end{bmatrix}, \quad \mathcal{D} = \begin{bmatrix} 0 \\ 0.1 \\ 0.1 \end{bmatrix}.$$

Fig. 8. State trajectories of $f_i(t)$ and $\hat{f}_i(t)$, $i \in \mathbb{F}$.

The Laplacian matrix of the communication topology is $\mathcal{L} = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$, and $\mathcal{J} = \text{diag}\{0, 1, 0, 0\}$. The time-varying delay of systems (1) is $d(t) = 0.1 |\sin(t)| + 0.1$ and the external disturbances is $w_i(t) = e^{-t} \sin(t)$.

From Theorem 4, the matrices $\mathcal{X}_1$, $\mathcal{X}_2$, and $\mathcal{K}$ have the following form:

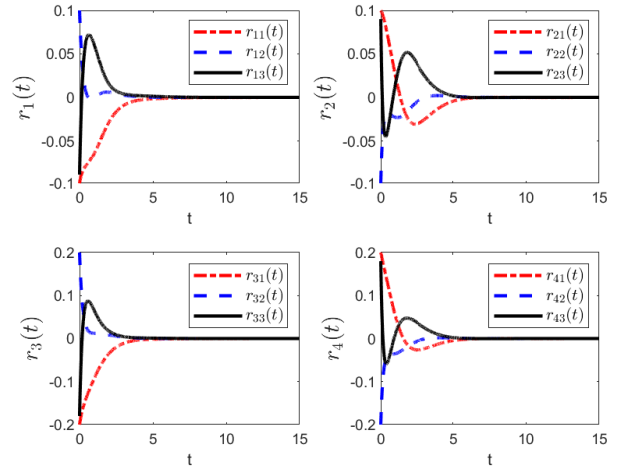
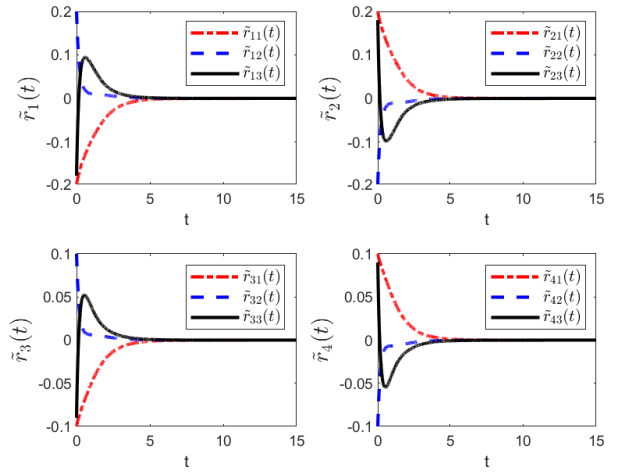
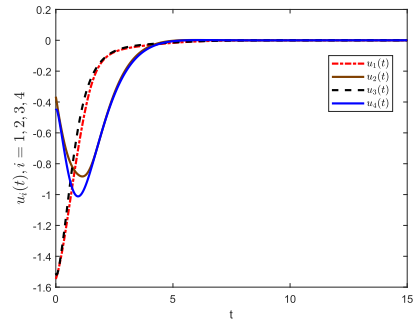
$$\mathcal{X}_1 = \begin{bmatrix} 1.0339 & 0 & -1.5664 \\ 0 & 1.0339 & -2.2008 \\ 0 & 0 & -0.9359 \end{bmatrix}$$

$$\mathcal{X}_2 = \begin{bmatrix} 1.1096 & 0 & -1.6267 \\ 0 & 1.1096 & -2.3306 \\ 0 & 0 & -0.9716 \end{bmatrix}$$

$$\mathcal{K} = -[0.1194 \quad 0.0457 \quad 0.1007].$$

The number of hidden layer nodes of the neural network is 10. Set $\hat{\Delta}_i(0) = 0.1I_{1,10}$, $\hat{\Pi}_i(0) = 0.1I_{7,10}$, and $\hat{\theta}_i(0) = 0.1I_{4,1}$ for all agent i , where all elements of $I_{p,q} \in \mathbb{R}^{p \times q}$ are 1. The positive matrix $\mathcal{P}_{1i} = 10\text{diag}(1, \dots, 1) \in \mathbb{R}^{10 \times 10}$, $\mathcal{P}_{2i} = 10\text{diag}(1, \dots, 1) \in \mathbb{R}^{7 \times 7}$, and $\mathcal{P}_{3i} = 10\text{diag}(1, 1, 1, 1)$ for $i = 1, 2, 3, 4$. The matrix $\mathcal{R} = I$, the initial conditions are chosen as $x_0(0) = [-1, 1, -0.9]^T$, $x_1(0) = [-1.1, 1.1, -0.99]^T$, $x_2(0) = [-0.9, 0.9, 0.81]^T$, $x_3(0) = [-1.2, 1.2, -1.08]^T$, and $x_4(0) = [-0.8, 0.8, -0.72]^T$.

Given $T = 15$ s, the singular multiagent systems (1) are FTCT respect to $(0.28, 12.8412, [0, T_r])$ and the error systems (14) are FTS respect to $(0.28, 11.9562, [0, T_r])$. Moreover, the FTCT of (1) and FTS of (14) over sliding motion phase within $[T_r, T]$ are true, where $l_2 = 301.4409$ and $\tilde{l}_2 = 280.8758$. The state trajectories of consensus tracking errors $r_1(t)$, $r_2(t)$, $r_3(t)$, and $r_4(t)$ are shown in Fig. 9. From Fig. 9, the consensus tracking errors $x_i(t) - x_0(t)$, $i \in \mathbb{F}$ are FTS, that is, the state of

Fig. 9. State trajectories of consensus tracking error $r_1(t)$, $r_2(t)$, $r_3(t)$, and $r_4(t)$.Fig. 10. State trajectories of $\tilde{r}_1(t)$, $\tilde{r}_2(t)$, $\tilde{r}_3(t)$, and $\tilde{r}_4(t)$.Fig. 11. State trajectories of $u_1(t)$, $u_2(t)$, $u_3(t)$, and $u_4(t)$.

follower agents $x_i(t)$, $i \in \mathbb{F}$ can track the state of leader agent $x_0(t)$ in finite time.

The state trajectories of $\tilde{r}_i(t)$, $i \in \mathbb{F}$, are displayed in Fig. 10. According to the simulation results in Fig. 10, the FTS of (14) over the interval $[0, 15]$ s is established. For all $i \in \mathbb{F}$ the stability of neural-adaptive consensus tracking protocol $u_i(t)$ is shown in Fig. 11. Select the nonsingular matrix $\mathcal{X} = 10I$,

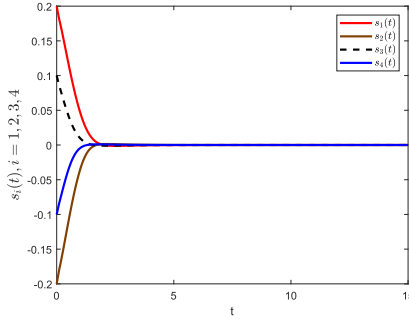


Fig. 12. State trajectories of $s_1(t)$, $s_2(t)$, $s_3(t)$, and $s_4(t)$.

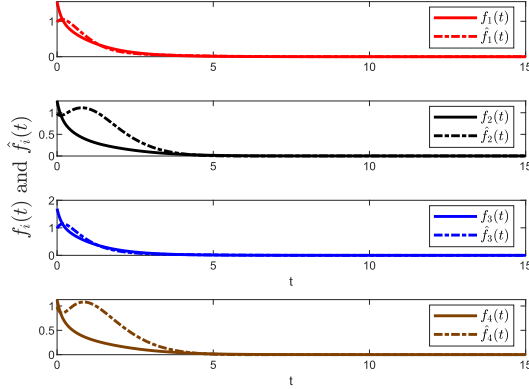


Fig. 13. State trajectories of $f_i(t)$ and $\hat{f}_i(t)$, $i = 1, 2, 3, 4$.

the sliding mode function in (10) is given as follows:

$$s_i(t) = [1 \ 2 \ 0] \tilde{r}_i(t) - \int_0^t ([6.5 \ 6.3 \ 9.4] \tilde{r}_i(\tau) + 0.01[1 \ 2 \ 0] \times \tilde{r}_i(\tau - d(\tau)) + [-0.0597 \ -0.0228 \ -0.0503] \times \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\tilde{r}_j(\tau) - \tilde{r}_i(\tau)) - J_i \tilde{r}_i(\tau) \right)) d\tau.$$

And the simulation results of state trajectories $s_i(t)$ for $i \in \mathbb{F}$ are displayed in Fig. 12. The sliding mode function is stable based on the results in Fig. 12. The state trajectories of $f_i(t)$ and $\hat{f}_i(t)$, $i \in \mathbb{F}$ are given in Fig. 13. From Fig. 13, the error $f_i - \hat{f}_i$ tends to zero, that is, the nonlinear actuator attacks f_i can be accurately estimated by the neural network method in this article.

V. CONCLUSION

Considering the nonlinear actuator attack, time-varying delay, and singular dynamics, the FTCT control problem analysis is more challenging in this article. In order to reduce the influence of actuator attack on the reachability of the sliding mode surface, the states observer of each follower agent are designed. The novel neural-adaptive consensus tracking protocol is designed by updating neural network parameters, which can drive the consensus tracking error to zero. Finally, the sufficient conditions for the FTCT of singular multiagent systems are given based on the sliding mode control with a partitioning strategy. The finite time formation of a singular multiagent is also an interesting direction worth further research.

REFERENCES

- [1] J. Yu and Y. Shi, "Scaled group consensus in multiagent systems with first/second-order continuous dynamics," *IEEE Trans. Cybern.*, vol. 48, no. 8, pp. 2259–2271, Aug. 2018.
- [2] L. Li, P. Shi, and C. K. Ahn, "Distributed iterative FIR consensus filter for multiagent systems over sensor networks," *IEEE Trans. Cybern.*, vol. 52, no. 6, pp. 4647–4660, Jun. 2022.
- [3] Q. Zhou, P. Shi, Y. Tian, and M. Wang, "Approximation-based adaptive tracking control for MIMO nonlinear systems with input saturation," *IEEE Trans. Cybern.*, vol. 45, no. 10, pp. 2119–2128, Oct. 2015.
- [4] S. Dong, W. Ren, Z. Wu, and H. Su, "∞ output consensus for Markov jump multiagent systems with uncertainties," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 2264–2273, May 2020.
- [5] X. Li, Y. Tang, Y. Zou, S. Li, and W. X. Zheng, "Event-triggered multi-agent consensus under relative output sensing," *IEEE Trans. Cybern.*, vol. 54, no. 2, pp. 915–928, Feb. 2024.
- [6] J. Yu, X. Dong, Q. Li, and Z. Ren, "Practical time-varying formation tracking for second-order nonlinear multiagent systems with multiple leaders using adaptive neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 6015–6025, Dec. 2018.
- [7] H. Li, P. Xie, and W. Yan, "Receding horizon formation tracking control of constrained underactuated autonomous underwater vehicles," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5004–5013, Jun. 2017.
- [8] Y. Liu, X. Shang, L. Tang, and S. Zhang, "Finite-time consensus adaptive neural network control for nonlinear multiagent systems under PDE models," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6218–6228, Apr. 2025.
- [9] X. Liu, X. Wu, Y. Xie, B. Guo, and J. Yan, "Fully distributed adaptive time-varying formation control of singular multiagent systems," *Neurocomputing*, vol. 516, pp. 146–154, Jan. 2023.
- [10] S. Li and X. Wang, "Finite-time consensus and collision avoidance control algorithms for multiple AUVs," *Automatica*, vol. 49, no. 11, pp. 3359–3367, Nov. 2013.
- [11] M. Zhao, C. Peng, and E. Tian, "Finite-time and fixed-time bipartite consensus tracking of multi-agent systems with weighted antagonistic interactions," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 1, pp. 426–433, Jan. 2021.
- [12] A. Nandanwar, N. K. Dhar, D. Malyshev, L. Rybak, and L. Behera, "Finite-time robust admissible consensus control of multirobot system under dynamic events," *IEEE Syst. J.*, vol. 15, no. 1, pp. 780–790, Mar. 2021.
- [13] R. Sakthivel, S. Kanakalakshmi, B. Kaviarasan, Y.-K. Ma, and A. Lee-lamani, "Finite-time consensus of input delayed multi-agent systems via non-fragile controller subject to switching topology," *Neurocomputing*, vol. 325, pp. 225–233, Jan. 2019.
- [14] B. Mu, K. Zhang, and Y. Shi, "Integral sliding mode flight controller design for a quadrotor and the application in a heterogeneous multi-agent system," *IEEE Trans. Ind. Electron.*, vol. 64, no. 12, pp. 9389–9398, Dec. 2017.
- [15] J. Song, Y. Niu, and Y. Zou, "Finite-time stabilization via sliding mode control," *IEEE Trans. Autom. Control*, vol. 62, no. 3, pp. 1478–1483, Mar. 2017.
- [16] Z. Ye, D. Zhang, C. Deng, H. Yan, and G. Feng, "Finite-time resilient sliding mode control of nonlinear UMV systems subject to DoS attacks," *Automatica*, vol. 156, Oct. 2023, Art. no. 111170.
- [17] R. Nie, W. Du, Z. Li, and S. He, "Sliding mode-based finite-time consensus tracking control for multi-agent systems under actuator attacks," *Inf. Sci.*, vol. 640, Sep. 2023, Art. no. 118971.
- [18] P. Zagalák, *Singular Control Systems*, vol. 28. Berlin, Germany: Springer, 1989, pp. 649–650.
- [19] B. Jiang, C. Gao, and J. Xie, "Passivity based sliding mode control of uncertain singular Markovian jump systems with time-varying delay and nonlinear perturbations," *Appl. Math. Comput.*, vol. 271, pp. 187–200, Nov. 2015.
- [20] J. Zhu, G. Zhuang, X. Xie, and J. Xia, "Distributed dual-terminal adaptive triggered consensus tracking of descriptor multi-agent systems against asynchronous DoS attacks with Markov switching topologies," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 12, pp. 6265–6278, Dec. 2024.
- [21] X.-R. Yang and G.-P. Liu, "Necessary and sufficient consensus conditions of descriptor multi-agent systems," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 59, no. 11, pp. 2669–2677, Nov. 2012.
- [22] J. Xi, Y. Yu, G. Liu, and Y. Zhong, "Guaranteed-cost consensus for singular multi-agent systems with switching topologies," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 61, no. 5, pp. 1531–1542, May 2014.

- [23] T. Liu, A. Sheng, G. Qi, and Y. Li, "Fully distributed bipartite output formation tracking for heterogeneous singular multiagent systems on signed digraphs," *Int. J. Robust Nonlinear Control*, vol. 33, no. 17, pp. 10698–10715, Nov. 2023.
- [24] P. Zhang, H. Xue, S. Gao, X. Zuo, and J. Zhang, "Finite-time adaptive cooperative fault-tolerant control for multi-agent system with hybrid actuator faults," *IEEE Syst. J.*, vol. 16, no. 3, pp. 3590–3601, Sep. 2022.
- [25] T. Yang, H. Kang, H. Ma, and X. Wang, "Adaptive fuzzy finite-time fault-tolerant consensus tracking control for high-order multiagent systems with directed graphs," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 607–616, Jan. 2023.
- [26] X. Liu, Y. Xie, F. Li, and W. Gui, "Consensus tracking of singular multiagent systems via sliding-mode approach," *IEEE Trans. Autom. Control*, vol. 68, no. 8, pp. 5022–5028, Aug. 2023.
- [27] Y. Zhang, Z. Zhao, and S. Ma, "Adaptive neural consensus tracking control for singular multi-agent systems against actuator attacks under jointly connected topologies," *Nonlinear Dyn.*, vol. 113, no. 4, pp. 3563–3577, Feb. 2025.
- [28] X. Jin, W. M. Haddad, and T. Yucelen, "An adaptive control architecture for mitigating sensor and actuator attacks in cyber-physical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 11, pp. 6058–6064, Nov. 2017.
- [29] R. Gao, J. Huang, and L. Wang, "Leaderless consensus control of uncertain multi-agents systems with sensor and actuator attacks," *Inf. Sci.*, vol. 505, pp. 144–156, Dec. 2019.
- [30] L. Cheng, Y. Wang, W. Ren, Z.-G. Hou, and M. Tan, "On convergence rate of leader-following consensus of linear multi-agent systems with communication noises," *IEEE Trans. Autom. Control*, vol. 61, no. 11, pp. 3586–3592, Nov. 2016.
- [31] Y. Niu, J. Lam, X. Wang, and D. W. C. Ho, "Neural adaptive sliding mode control for a class of nonlinear neutral delay systems," *J. Dyn. Syst., Meas., Control*, vol. 130, no. 6, Nov. 2008, Art. no. 061011.



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